ShareMate: A Socially Connected Smart Mobility and Community Support App

SHARED GROCERY SHOPPING AND DELIVERY MODULE

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DECLARATION

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ABSTRACT

ShareMate is a community-based platform designed to simplify everyday tasks for university students through collaborative resource sharing. This research focuses on the Shared Grocery Shopping and Delivery Module, a system that enables students to request grocery items from trusted community members who are already present at nearby supermarkets. By leveraging peer-to-peer collaboration, the module reduces the time, effort, and transportation barriers that students face when purchasing essential food items.

A key feature of the module is the supermarket-based shopper discovery mechanism. A requester first selects a preferred supermarket and is then shown a list of available shoppers who are currently present at that location. To support informed decision-making, each shopper profile presents delivery counts, ratings, feedback scores, and a trust-based performance graph that categorizes shopper reliability into the levels Excellent, Very Good, and Good.

The module also introduces an alternative item selection mechanism, allowing requesters to nominate substitute products in case a requested item is out of stock, which reduces request failures. A multi-request handling engine lets a single shopper serve several requesters at once while maintaining a separate, itemized bill for each requester. After the final cost, including product prices and delivery charges, is calculated, the shopper and requesters meet at a designated location within the university premises to complete the handover and payment.

The system follows a three-layer architecture (presentation, application and data layers) built with a responsive Flutter frontend, Python backend and a database layer with real-time updates. The module was verified through unit, integration, performance and usability testing. Overall, the Shared Grocery Shopping and Delivery Module enhances campus collaboration by combining real-time shopper availability, trust-based shopper evaluation, flexible item substitution and transparent cost management, offering a practical, reliable and socially driven solution that improves convenience and accessibility within the university environment.

Keywords: *Peer-to-Peer Shopping; Community Collaboration; Shopper Discovery; Trust-Based Evaluation*

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LIST OF ABBREVIATIONS

Abbreviation	Description
API	Application Programming Interface
ASVS	Application Security Verification Standard
CSS	Cascading Style Sheets
DB	Database
ER	Entity-Relationship
GPS	Global Positioning System
HTML	HyperText Markup Language
IT	Information Technology
JS	JavaScript
JSON	JavaScript Object Notation
LKR	Sri Lankan Rupees
MoSCoW	Must, Should, Could, Won't
OWASP	Open Worldwide Application Security Project
P2P	Peer-to-Peer
PDPA	Personal Data Protection Act
REST	Representational State Transfer
RQ	Research Question
SDG	Sustainable Development Goal
SUS	System Usability Scale
UAT	User Acceptance Testing
UI	User Interface
UX	User Experience
WBS	Work Breakdown Structure
WCAG	Web Content Accessibility Guidelines

1 INTRODUCTION

This chapter introduces the Shared Grocery Shopping and Delivery module of the ShareMate application. It places the module within the overall project, defines the problem and the research gap, states the aim, objectives, research questions and hypotheses, and clarifies the scope and the author's individual contribution. The chapter ends with an overview of the structure of the thesis.

1.1 Overall project context

ShareMate: A Socially Connected Smart Mobility and Community Support App is a community-based digital platform developed to support everyday mobility, academic, and service-related needs within university student communities. The platform is based on the premise that many resources already available within a student community, including travel capacity, local knowledge, time, academic materials, and willingness to assist others, remain under-utilised because students lack an integrated mechanism for identifying, coordinating, and trusting potential sources of support. Rather than relying exclusively on commercial service providers or isolated applications for individual needs, ShareMate enables students to participate in a connected environment where community members can request, offer, and coordinate services with one another. This approach is consistent with research on the sharing economy and collaborative consumption, which demonstrates how digital platforms can facilitate the use of under-utilised resources through peer-to-peer interaction and community participation [1], [7], [12].

The motivation for ShareMate arises from the interconnected nature of student life. University students frequently encounter challenges involving transportation, travel coordination, academic resources, and everyday assistance. These challenges are often addressed through separate applications or informal communication channels, such as messaging groups and individual arrangements. Although such approaches can provide immediate solutions, they provide limited support for structured matching, availability management, location-based coordination, trust establishment, and maintaining a consistent record of community interactions. ShareMate therefore explores whether these different requirements can be brought together within a single community-oriented platform while retaining separate mechanisms for each type of task.

The platform is developed as a group research project in which different team members are responsible for individual functional components. Although each component addresses a distinct problem, the components operate within a shared application environment and follow a common community-oriented approach. Shared platform capabilities include user accounts, authentication, notifications, and community profiles, while the individual components provide domain-specific functionality. This structure allows ShareMate to be viewed not simply as a collection of independent features, but as a connected platform in which identity, community participation, location, availability, and interaction history can support different forms of peer-to-peer coordination. Figure 1 illustrates the position of the component investigated in this study within the wider ShareMate platform.

The mobility-oriented functionality of ShareMate focuses on reducing the coordination difficulties associated with student transportation. The FriendZone Ride component allows students with similar travel requirements to identify potential ride-sharing opportunities based on relevant information such as destination, travel timing, and community relationships. The platform also incorporates fare-related functionality to support the calculation and coordination of shared travel costs. In addition, the FLTS component provides location-aware public transport support by using positioning and mapping information to provide bus-related tracking and estimated arrival information. These functions address different aspects of student mobility: one focuses on peer-to-peer travel coordination, while the other supports access to transport information.

ShareMate also extends the community-sharing concept beyond transportation. The Grocery Coordination component is intended to support situations in which a student requires an item from a supermarket or similar location while another community member is already visiting or is willing to make the trip. Instead of requiring each request to generate an independent journey, the component provides a mechanism for connecting requests with potential shoppers and coordinating fulfilment. This represents a community-based form of service sharing in which existing movement and availability can potentially be used to satisfy another student's requirement. The coordination process incorporates factors such as request information, shopper availability, matching, and completion of the request.

The academic-support aspect of the platform is represented by the Academic Notes and Information/Recommendation component (ANIE), which provides functionality for organising and recommending academic materials according to the requirements of students. This component extends the resource-sharing principle from physical and mobility resources to

information resources. Students can therefore interact with the platform not only to coordinate travel or everyday services, but also to access academic content relevant to their needs. The combination of these functions reflects the broader objective of ShareMate: to provide a connected community environment in which different types of student requirements can be addressed through structured peer interaction.

A key characteristic of ShareMate is that the different components involve different types of data, decision processes, and evaluation requirements. Mobility-related functions depend on location and temporal information, ride matching depends on compatibility between users and travel requirements, grocery coordination depends on request and shopper availability, and academic recommendation depends on the characteristics and relevance of available learning resources. Consequently, the platform cannot be evaluated appropriately using a single performance measure. Each component requires task-specific evaluation criteria that reflect the nature of the underlying problem. This distinction is important because a percentage obtained from a classification task, for example, cannot be interpreted as equivalent to a percentage obtained from GPS performance, ride matching, recommendation, or grocery completion.

The research therefore evaluates ShareMate through multiple task-specific measures corresponding to the functions investigated in the study. The evaluation considers the performance of FLTS in terms of ETA and location-related outcomes, ANIE in terms of note classification and recommendation outcomes, FriendZone Ride in terms of matching and fare correctness, and Grocery Coordination in terms of shopper matching and request completion. These measures provide a component-level view of the platform while recognising that the underlying tasks are heterogeneous. Rather than combining these measurements into an artificial overall accuracy value, the results are interpreted according to the specific purpose and evaluation criteria of each component.

From a systems perspective, ShareMate also investigates the practical implications of integrating multiple community services within a common platform. The application uses a shared infrastructure for user and community interaction while allowing individual services to implement domain-specific processing. This provides an opportunity to examine how common identity and community information can support different coordination mechanisms, while also highlighting the dependencies introduced by shared services. For example, location-based functionality may depend on mapping and positioning services, while community functions require persistent user and interaction information. These dependencies are considered when interpreting the reliability, responsiveness, and practical applicability of the platform.

Trust and community participation are also important considerations in the ShareMate model. Peer-to-peer coordination requires users to make decisions about whom they are willing to interact with, particularly when activities involve shared travel, purchases, or repeated community interactions. User profiles, interaction history, and feedback mechanisms are therefore incorporated into the wider platform concept to provide contextual information for community-based coordination. However, such mechanisms do not automatically guarantee trustworthy behaviour. Their effectiveness depends on the availability and quality of participation data, the accuracy of user-provided information, and the continued participation of community members. These considerations are treated as part of the practical and governance implications of the platform rather than as assumptions of guaranteed trust.

The overall objective of ShareMate is consequently not to replace existing commercial mobility, delivery, or academic platforms, but to investigate how a university-centred digital community can coordinate resources that already exist among its members. By bringing mobility assistance, ride sharing, grocery coordination, and academic resource support into a connected environment, the platform explores a broader model of community-based digital assistance. The research contribution lies in the design and evaluation of this integrated approach and in examining how different coordination mechanisms perform across heterogeneous student-oriented tasks.

Within this wider ShareMate platform, this study focuses on the components and evaluation objectives defined in the present research. The subsequent sections describe the system architecture, individual components, implementation approach, evaluation methodology, and experimental results. The study evaluates each function using its appropriate task-specific measures and discusses the resulting performance in relation to the research questions, previous work, practical constraints, and limitations of the evaluation setting.

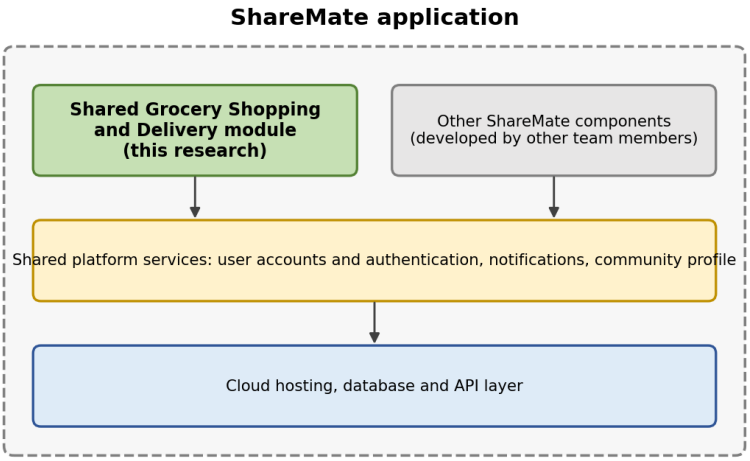


Figure 1: Position of the Shared Grocery Shopping and Delivery module within the ShareMate application

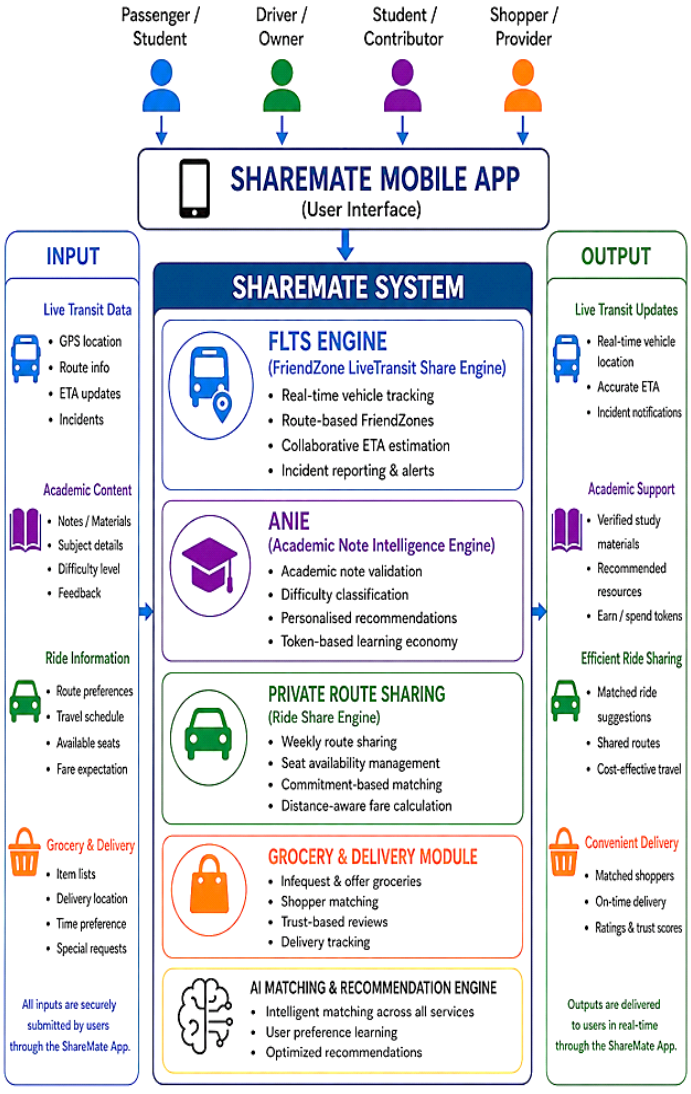


Figure 1.1: Overall ShareMate System Diagram

1.2 Individual component context

The Shared Grocery Shopping and Delivery module allows a university student (the requester) to ask a trusted fellow student (the shopper) who is already at a nearby supermarket to buy grocery items on their behalf. The shopper may serve several requesters during the same visit, keeps a separate bill for each of them, and hands the groceries over at an agreed location inside the university premises, where payment is settled.

Two user roles are therefore central. A **requester** selects a supermarket, views the shoppers who are currently present there, studies their trust profiles, chooses one, and submits a grocery list that may include alternative products. A **shopper** declares availability at a supermarket, receives and accepts requests, buys the items, records the actual prices, selects a transport method for the delivery, completes the handover and receives a rating. Any registered student can act in either role.

The module combines six functional ideas: supermarket-based shopper discovery, trust-based shopper evaluation, alternative item selection, multiple request handling by a single shopper, transparent cost calculation with separate bills, and delivery coordination within the university.

1.3 Problem statement

University students often find it difficult to manage everyday responsibilities such as grocery shopping because of academic workloads, limited free time and transportation constraints. Commercial grocery delivery services exist, but they rely on centralized delivery fleets, which can involve higher delivery costs, longer waiting times and limited flexibility [4], [8]. These platforms also ignore a valuable local resource: community members who are physically present at the same supermarket and could shop for others at almost no extra effort.

The research problem addressed in this study is therefore: **how can a trustworthy, flexible and transparent peer-to-peer grocery shopping and delivery service be designed for a university community, so that students can obtain groceries from peers who are already at supermarkets?** The problem has the following components.

1. **No community-based grocery sharing.** Existing systems depend on professional delivery services, so the opportunity to use people who are already at supermarkets is unexplored.
2. **No real-time identification of available shoppers.** Requesters cannot find individuals who are physically at a supermarket and willing to buy for others.

3. **Insufficient trust information.** Basic star ratings give little insight into how reliable a shopper really is, which reduces confidence in peer-to-peer services [2], [5], [11].
4. **Item unavailability.** When a requested item is out of stock, orders remain incomplete or require extra communication, delaying completion.
5. **Multiple requests and cost transparency.** When one shopper buys for several people, existing tools do not keep separate bills or share delivery charges transparently.
6. **Local delivery coordination.** Platforms designed for large-scale logistics do not support simple handovers and direct payments within a campus.

1.4 Research gap

A review of the literature and of existing applications (Chapter 2) shows that digital platforms for online grocery shopping and delivery mainly connect customers with supermarkets or professional couriers. Studies of the sharing economy and crowd-based work describe the benefits of peer-to-peer cooperation in general [1], [3], [9], and studies of reputation systems explain how trust can be supported [2], [5], [11], but little work combines these ideas for in-store grocery shopping inside a controlled community such as a university. Table 1 compares the proposed module with the foundational works reviewed.

Table 1: Comparison of the proposed module with existing research

Feature	**Sundararajan [1]**	**Jøsang et al. [2]**	**Gevaers et al. [4]**	**Resnick et al. [11]**	**Proposed module**
Community-based grocery shopping	✗	✗	✗	✗	✓
Real-time shopper availability in supermarkets	✗	✗	✗	✗	✓
Trust-based shopper performance evaluation	✗	✓	✗	✓	✓
Alternative item selection mechanism	✗	✗	✗	✗	✓
Transparent cost calculation and billing	✗	✗	✗	✗	✓
Multiple request handling by one shopper	✗	✗	✗	✗	✓
Delivery coordination within a university	✗	✗	✗	✗	✓

The gaps are summarized as follows. First, there is no accepted design for **community-based grocery shopping** that uses people already at the store. Second, **real-time shopper availability** by supermarket is not supported. Third, trust is usually shown as a single average rating, whereas a **graphical, category-based trust profile** could convey more. Fourth, **substitute handling** is rarely built into peer-to-peer requests. Fifth, **multi-request billing** with a separate bill for each requester is not addressed. Sixth, **campus-scale delivery coordination** with handover and payment at a designated place has not been studied. The proposed module addresses all six gaps in one integrated design.

1.5 Aim and objectives

1.5.1 Aim

The aim of this research is to design, develop and evaluate a community-based shared grocery shopping and delivery module within the ShareMate application that enables university students to request grocery items from trusted peers who are already present at supermarkets, improving convenience, efficiency and collaboration within the university community.

1.5.2 Objectives

To achieve the aim, the following specific objectives were set.

1. **O1 Shopper discovery.** Develop a supermarket-based discovery system that lists the shoppers currently present at a selected supermarket.
2. **O2 Trust evaluation.** Design a trust-based evaluation system that displays delivery counts, ratings, feedback scores and a graphical performance classification (Excellent, Very Good, Good).
3. **O3 Request and substitutes.** Implement a grocery request mechanism with optional alternative items for products that may be unavailable.
4. **O4 Multiple requests.** Support a single shopper handling several requests at the same time, with organized shopping and separate bills.
5. **O5 Cost transparency.** Provide transparent calculation of the final cost of each request, including item prices and delivery charges.
6. **O6 Delivery coordination.** Facilitate the handover of groceries and payment at designated locations within the university premises.
7. **O7 Evaluation.** Verify the module through functional, integration, performance and usability testing with students from the university community.

Table 2: Mapping of objectives, deliverables and evaluation measures

Objective	Main deliverable	Evaluation measure
O1	Supermarket selection and available-shopper list	Correctness of listed shoppers; list response time
O2	Shopper profile and trust performance graph	Correct computation of metrics; user understanding of profile
O3	Grocery list form with alternative items	Substitute offered when item unavailable; request success rate
O4	Shopper dashboard and merged checklist	Correct merging; separate allocation of items
O5	Itemized bill per requester	Bill accuracy (items + delivery charge)
O6	Handover confirmation and rating	Completion of full request lifecycle
O7	Test plan, test cases and usability study	Test pass rate; SUS score; user feedback

1.6 Research questions or hypotheses

The study is guided by the following research questions.

- **RQ1.** How can shoppers who are currently present at a supermarket be identified and presented to requesters in a reliable and timely way?
- **RQ2.** Does a category-based trust performance graph help requesters choose shoppers more confidently than a single average rating?
- **RQ3.** Does an alternative item mechanism reduce failed or incomplete grocery requests?
- **RQ4.** Can a single shopper handle several requests with separate bills while keeping costs clear for every requester?
- **RQ5.** Is the resulting module usable and acceptable to university students?

In addition, the following hypotheses are tested in the evaluation (Chapter 6).

- **H1.** Requesters who see the trust performance graph report higher confidence in their shopper choice than when only an average rating is shown.
- **H2.** Requests that include alternative items are completed more often than requests without alternatives when stock is limited.
- **H3.** The module computes every requester's bill correctly when a shopper serves multiple requests.
- **H4.** The module achieves a System Usability Scale score at or above the commonly cited average of 68 [15].

1.7 Scope limitations and assumptions

Table 3: Scope, limitations and assumptions

Aspect	Details
In scope	Supermarket selection; available-shopper discovery; shopper trust profiles; grocery lists with alternative items; multi-request handling; separate itemized bills; delivery charge calculation; handover coordination within the university; ratings and feedback; responsive web interface.
Out of scope	Online payment gateways and wallets; integration with supermarket inventory systems; delivery outside the university; route optimization; handling of age-restricted goods; the other ShareMate components.
Limitations	Evaluation limited to one university community and a small number of supermarkets; shopper presence relies on device location and user honesty; trust scores depend on enough completed deliveries.
Assumptions	Users are registered university students with smartphones and internet access; shoppers are willing to shop for others voluntarily; payment is settled in cash or by direct transfer at handover; a designated handover location exists on campus.

1.8 Individual contribution

This thesis reports the individual work of the author (Dayananda STPCT, IT22298058) on the Shared Grocery Shopping and Delivery module. The contributions are listed in Table 4.

Table 4: Individual contributions of the author

Area	Individual contribution
Research	Problem analysis, literature review, identification of research gaps, definition of objectives, research questions and hypotheses.
Requirements	Design and administration of the student survey; derivation of functional and non-functional requirements.
Design	System architecture, data model, request lifecycle, trust evaluation model, substitution logic, billing logic and user interface prototypes.
Implementation	Front end, back end and database of the module; real-time availability and notification features.
Testing and evaluation	Test plan and test cases; performance checks; usability study and analysis of feedback.
Documentation	Project proposal, this final report and supporting appendices.

1.9 Thesis structure

The thesis is organized into nine chapters. Chapter 1 introduces the study. Chapter 2 reviews the literature and existing systems and derives the conceptual framework. Chapter 3 describes the research methodology, including the research design, participants, procedure, measures, analysis, ethics and reproducibility. Chapter 4 presents the product and solution design, including requirements, architecture, design decisions and detailed design. Chapter 5 describes the implementation, integration and testing. Chapter 6 reports the results and validation. Chapter 7 discusses the findings, contributions and limitations. Chapter 8 considers commercialization and sustainability. Chapter 9 concludes the thesis and proposes future directions. References, a glossary and appendices follow.

2 LITERATURE REVIEW AND RELATED WORK

This chapter reviews the literature and the existing systems relevant to community-based grocery shopping. It explains how the review was carried out, presents the theoretical foundations, summarizes related research, compares existing products, and synthesizes the gaps into a conceptual framework that guides the rest of the study.

2.1 Review method and scope

The grocery coordination component of ShareMate is positioned at the intersection of the sharing economy, peer-to-peer service coordination, crowdsourced delivery, and last-mile logistics. The underlying concept is that a student who is already travelling to a supermarket or who is willing to make a shopping trip can potentially purchase and deliver items requested by another member of the same community. This differs from conventional grocery delivery, where a dedicated employee, courier, or third-party delivery service is responsible for fulfilling each order. The ShareMate approach instead attempts to utilise existing community participation and travel activity to coordinate grocery requests with available shoppers.

Research on the sharing economy provides the broader theoretical foundation for this approach. Sharing-economy platforms enable individuals to obtain, provide, or share access to goods and services through digitally mediated peer-to-peer interactions. The value of such systems is partly derived from making previously under-utilised resources available to other participants. In the context of grocery coordination, the relevant resource is not limited to physical goods; it can also include a community member's time, shopping capacity, and existing movement between locations. This creates a potential connection between individual shopping activity and the fulfilment of another user's demand.

Crowdsourced delivery extends this principle into last-mile logistics by using members of a crowd to perform delivery activities. Alnaggar, Gzara, and Bookbinder [26] identify crowdsourced delivery as an emerging last-mile model and classify systems according to factors such as matching mechanisms, target markets, and compensation schemes. Their review also highlights that crowdsourced delivery introduces operational challenges that differ from conventional delivery systems, including the availability of non-professional couriers, matching of orders with couriers, scheduling, and uncertainty in delivery operations. These characteristics are particularly relevant to ShareMate because the proposed grocery shopper is not assumed to be a professional delivery worker. Instead, the shopper is a member of the university community whose availability is linked to personal activities and circumstances.

A particularly relevant stream of research investigates **in-store customers as crowdsourced delivery participants**. Devari et al. [27] examined the possibility of crowdsourcing the last mile by exploiting the social networks of retail-store customers. This work is important to the ShareMate grocery component because it demonstrates how customers who are already visiting a retail location can potentially contribute to the fulfilment of other customers' online orders. The mechanism therefore differs from a traditional courier model: the delivery capacity is generated by people whose primary reason for travelling is their own shopping activity rather than the delivery itself.

More recent research provides an even closer operational comparison. De Maio et al. [28] investigate an in-store crowdshipping system in which customers who are checking out of a store can declare themselves available to deliver one or more requests from online shoppers. Their model considers a dispatching platform that combines dedicated delivery capacity with crowdsourced couriers and addresses the routing of store pickups and subsequent deliveries. This work demonstrates that in-store crowdshipping is not simply a conceptual extension of conventional delivery; it introduces dynamic decisions regarding which available customer should be assigned to which request and how multiple orders should be incorporated into the delivery process. These characteristics are directly relevant to the ShareMate grocery coordination mechanism, where a student shopper can potentially be associated with one or more requests based on availability and compatibility.

The literature also demonstrates that crowdsourced delivery involves a significant **matching problem**. A delivery request cannot be considered fulfilled merely because a potential courier exists. The platform must identify an appropriate match based on factors such as location, timing, capacity, route compatibility, and willingness to participate. Alnaggar et al. [26] show that matching and scheduling mechanisms are central characteristics by which crowdsourced delivery systems differ. Research on crowdsourced delivery also identifies the importance of spatio-temporal compatibility between delivery requests and the routes or locations of participating couriers. This suggests that grocery coordination should consider not only whether a shopper is available, but also whether the shopper's location and planned movement make the fulfilment of a request feasible.

The importance of uncertainty is particularly relevant when the crowd consists of ordinary users rather than professional couriers. A shopper's availability may change, a shopping trip may take longer than expected, an item may not be available in the store, or the shopper may decide not to complete the additional delivery. Recent research on crowdsourced delivery

planning identifies uncertainty in task duration, courier availability, and order assignment as continuing operational challenges [29]. Consequently, a community grocery platform requires mechanisms for handling cancellations, unmatched requests, and unsuccessful completion rather than assuming that every accepted assignment will result in a completed delivery.

The grocery-specific nature of the problem also distinguishes ShareMate from generic peer-to-peer delivery. Grocery requests can contain multiple items, may depend on store availability, and can involve different pickup and drop-off requirements. A shopper may already be travelling to a supermarket, making the marginal effort of purchasing another user's items potentially different from generating a dedicated delivery journey. Recent work on peer-to-peer crowdshipping similarly examines the strategic value of using in-store customers to support online order fulfilment [30]. Such studies provide support for the underlying principle of ShareMate, while also demonstrating that the benefits of this model depend on how customers, orders, routes, and incentives are coordinated.

Another important dimension is **trust**. Unlike a conventional supermarket delivery service, a peer-to-peer grocery transaction involves interaction between members of a community who may not know each other personally. The requesting student must therefore have sufficient confidence that another student will purchase the requested items, handle them appropriately, and complete the delivery as agreed. Trust is recognised as an important factor in the adoption of crowdsourced delivery services. Cebeci et al. [31] found that trust influences the choice of crowdshipping services and reported that trust can mediate the relationship between service attributes and adoption. Their findings indicate that factors associated with reputation and potential delivery damage can influence adoption through trust.

The broader sharing-economy literature similarly identifies reputation systems, ratings, reviews, identity information, and platform-level mechanisms as important approaches to reducing uncertainty between participants. Räisänen, Ojala, and Tuovinen [32], in a systematic review of trust in the sharing economy, identify multiple technological mechanisms for building trust, with reputation systems, ratings, and reviews among the commonly used approaches. However, trust mechanisms should not be interpreted as guarantees of reliability. Research on user-generated reputation information has shown that users can exhibit positive biases when interpreting peer reputation information [33]. This is relevant to ShareMate because a high community rating may provide useful contextual information about a shopper, but it does not necessarily establish that the individual will be reliable for every future grocery transaction.

Identity and transparency can also influence perceptions of peer-to-peer delivery. Research on crowdsourced delivery design has shown that disclosure of information about the delivery participant can influence customer trust and satisfaction [33]. For a university-centred system such as ShareMate, the use of verified student identities and community profiles can therefore provide contextual information that is not normally available in an anonymous transaction. Nevertheless, the collection and presentation of identity and reputation information must be balanced against privacy and data-minimisation requirements.

The literature therefore suggests that a successful peer-to-peer grocery coordination system requires more than a basic request-and-response mechanism. At minimum, it must address the relationship between **request characteristics, shopper availability, location, matching, trust, and completion**. These factors are interdependent. For example, increasing the number of potential shoppers may improve the opportunity for matching, but it may also increase the complexity of selecting an appropriate participant. Similarly, a reputation mechanism may provide additional information for matching or user decision-making, but sparse or biased ratings may provide an unreliable representation of actual performance.

Research Gap and Positioning of ShareMate

Although existing research provides substantial evidence for crowdsourced and in-store delivery models, much of the literature focuses on commercial last-mile logistics, urban delivery networks, optimisation models, or dedicated crowdsourced delivery platforms. The assumptions underlying these systems do not necessarily correspond to a small, university-centred community in which participants have overlapping social relationships, recurring schedules, and non-commercial motivations.

ShareMate addresses this context by considering grocery coordination as a **community-level peer-to-peer service**. Instead of treating the shopper exclusively as a courier, the proposed mechanism considers the shopper as a member of the same community who may already have a reason to visit a supermarket. The system consequently focuses on coordinating existing community availability with grocery requests rather than constructing an independent professional delivery workforce.

The proposed mechanism also incorporates community identity and interaction information into the coordination process. This provides a basis for considering trust and previous interaction history alongside conventional matching factors. However, these mechanisms are treated as coordination aids rather than evidence that every matched shopper will successfully

complete an order. The distinction is important because a successful match and a successful grocery completion represent different stages of the process and should therefore be evaluated separately.

Accordingly, the ShareMate grocery component evaluates **shopper matching and request completion as separate outcomes**. Matching measures whether a suitable shopper-request association is established, whereas completion measures whether the matched request is subsequently fulfilled. This distinction follows the operational logic identified in the crowdsourced delivery literature and avoids treating an assignment as equivalent to a completed delivery. It also provides a basis for analysing unsuccessful cases, including unmatched requests, cancellations, and incomplete orders.

Overall, the literature supports the feasibility of using community participants and in-store shoppers as an alternative source of last-mile delivery capacity, while also highlighting the challenges associated with dynamic availability, spatio-temporal matching, uncertainty, trust, and successful completion. ShareMate builds on these findings by applying the underlying crowdshipping principle to a university community and by integrating grocery requests, shopper participation, identity, and community interaction within a broader student-oriented platform. The resulting research gap is therefore not the invention of crowdsourced delivery itself, but the investigation of how a community-based grocery coordination mechanism can be structured and evaluated within a university environment where the participants, incentives, trust relationships, and usage patterns differ from those of commercial last-mile delivery systems.

Table 5: Literature review scope and search strategy

Item	Enhanced description
Topics	Sharing economy and collaborative consumption; peer-to-peer service platforms; crowdshipping and crowdsourced delivery; in-store customer delivery; peer-to-peer grocery delivery; grocery order fulfilment; last-mile delivery; shopper–request matching; spatio-temporal matching; courier availability and uncertainty; trust and reputation systems; identity verification; delivery completion and cancellation; community-based service coordination.
Search terms	"sharing economy", "collaborative consumption", "crowdshipping", "crowdsourced delivery", "peer-to-peer delivery", "peer-to-peer grocery delivery", "in-store customers delivery", "in-store crowdshipping", "grocery delivery", "crowd logistics", "last-mile delivery", "shopper matching", "courier-order matching", "delivery trust", "crowdshipping trust", "reputation system", "peer-to-peer trust", "delivery completion", "delivery cancellation".
Sources	Peer-reviewed journals and conference papers indexed in IEEE Xplore, ACM Digital Library, Scopus, Web of Science and Google Scholar; relevant books, standards and authoritative technical documentation where appropriate.
Inclusion criteria	Studies directly addressing sharing/collaborative consumption, crowdsourced or peer-to-peer delivery, grocery or in-store delivery, matching and routing, trust/reputation, or evaluation of comparable platforms; peer-reviewed or otherwise academically verifiable sources; studies with a clear methodological or conceptual contribution.
Exclusion criteria	Marketing material without supporting evidence; non-verifiable sources; papers addressing conventional delivery without relevance to crowd-based or peer-to-peer coordination; duplicate publications; sources whose findings cannot reasonably be related to the ShareMate research questions.
Period	Foundational research from 2000 onwards, with emphasis on recent research from 2021–2026 for crowdsourced delivery, trust, matching and peer-to-peer service coordination.

2.2 Domain and theoretical foundations

2.2.1 Sharing economy and collaborative consumption

Sundararajan [1] describes the sharing economy as a shift towards crowd-based capitalism, in which digital platforms coordinate the supply of goods and services from distributed individuals rather than from large firms. Botsman and Rogers [7] describe collaborative consumption as the sharing, exchanging and renting of resources without ownership, enabled by technology. Hamari et al. [12] found that motivations for taking part in collaborative consumption include sustainability, enjoyment and economic benefit, which is relevant for students who are cost-conscious and social. Benkler [9] argues that commons-based peer production can coordinate effort efficiently without conventional firm structures, which supports the idea that community members can organize a service among themselves.

Brynjolfsson and McAfee [10] discuss how digital technologies change the way work and services are organized.

Together these works justify the central design premise of the module: students who are already at a supermarket possess spare capacity (their trip and their time in the shop) that can be shared at low marginal cost.

2.2.2 Trust and reputation in peer-to-peer services

When people rely on strangers, trust becomes the critical enabling factor. Resnick and Zeckhauser [13] studied trust among strangers in internet transactions and showed that reputation feedback encourages cooperative behaviour. Resnick et al. [11] and Dellarocas [5] discuss the promise and challenges of online reputation systems, including issues such as rating inflation, reluctance to give negative feedback and the difficulty of newcomers. Jøsang et al. [2] survey trust and reputation systems for online service provision and describe how aggregated scores, probabilistic models and visual summaries support decision making. A common finding is that richer, more transparent reputation information helps users judge reliability better than a single number.

These works motivate the trust evaluation model of the module: it displays not only the average rating but also the number of completed deliveries, the feedback score and the distribution of deliveries over the categories Excellent, Very Good and Good.

2.2.3 Crowdsourcing and crowd-based work

Brabham [3] defines crowdsourcing as an online, distributed problem-solving and production model in which tasks are outsourced to a crowd. In a grocery context, each shopping errand is a micro-task that a nearby member of the crowd performs for a requester. Crowdsourced tasks work best when they are well specified, easy to accept and easy to verify, which motivates the structured grocery list, substitute rules and itemized bills used in this module.

2.2.4 Last-mile logistics

Gevaers et al. [4] analyse the characteristics and challenges of last-mile logistics for e-commerce and conclude that the last mile is often the most expensive and least efficient part of the delivery chain, particularly because of failed deliveries and low drop density. Large

operators describe their own last-mile systems [8], which rely on dedicated fleets and sophisticated routing. In a campus, the drop density is high and the delivery distance is short, so a handover at a single designated location can remove most of the last-mile cost.

2.2.5 Data-intensive application design

Kleppmann [6] discusses reliability, scalability and maintainability as the central concerns of data-intensive applications, along with data models, replication and consistency. These concerns apply directly to the module: shopper availability must be fresh, bills must be consistent, and the system must scale as more students join. They shaped the layered architecture and the choice of a real-time database described in Chapter 4.

2.2.6 Usability and software quality

Usability engineering [21] emphasizes learnability, efficiency, error prevention and satisfaction. The System Usability Scale [15] provides a quick, widely used measure of perceived usability. The ISO/IEC 25010 product quality model [16] defines characteristics such as functional suitability, performance efficiency, usability, reliability, security and maintainability, and is used in this study to structure the non-functional requirements and the evaluation.

2.3 Related research

Table 6 summarizes the main related works, their contribution and their limitation relative to the problem studied here.

Table 6: Summary of related research and its limitations relative to this study

Ref.	Focus	Contribution	Limitation for this study
[1] Sundararajan	Sharing economy	Explains crowd-based capitalism and platform-mediated exchange.	General; no design for in-store shopping or campus use.
[2] Jøsang et al.	Trust and reputation systems	Survey of trust models and reputation mechanisms.	Not specific to grocery shopping; no delivery coordination.
[3] Brabham	Crowdsourcing	Defines crowdsourcing as a problem-solving model.	Focus on knowledge tasks, not physical errands.
[4] Gevaers et al.	Last-mile logistics	Characterizes costs and challenges of last-mile delivery.	Assumes professional carriers; no peer delivery.

Ref.	Focus	Contribution	Limitation for this study
[5] Dellarocas	Online reputation	Promise and challenges of reputation systems.	Does not cover category-based performance graphs.
[6] Kleppmann	Data systems	Principles for reliable, scalable data systems.	General engineering guidance.
[7] Botsman and Rogers	Collaborative consumption	Describes sharing models and their social value.	No technical design.
[8] Amazon	Last-mile systems	Describes large-scale delivery logistics.	Commercial fleet model, not peer-based.
[9] Benkler	Peer production	Shows efficiency of commons-based cooperation.	Software and information goods rather than errands.
[10] Brynjolfsson and McAfee	Digital economy	Discusses technological change in work and services.	No specific platform design.
[11] Resnick et al.	Reputation systems	Foundational discussion of reputation systems.	Predates mobile, location-based peer services.
[12] Hamari et al.	Participation motives	Empirical study of why people join collaborative consumption.	Does not propose a system.
[13] Resnick and Zeckhauser	Trust among strangers	Evidence that feedback fosters trust.	Focus on online auctions.

2.4 Existing systems products and techniques

Several kinds of existing solutions can meet a student's grocery needs. They are described here in general terms based on public information, and Table 7 compares them with the proposed module.

- **Supermarket online ordering and delivery.** Large supermarket chains in Sri Lanka offer online ordering with home delivery. They provide a wide product range and reliable stock information, but delivery is performed by the retailer or its contractors, delivery fees and time slots apply, and there is no peer cooperation.
- **Food and courier delivery apps.** Mobile apps such as PickMe or Uber Eats connect customers with riders. They focus on restaurant food or parcels and rely on paid riders rather than on students who are already at the store.
- **Personal-shopper marketplaces.** Services in some countries allow paid shoppers to buy groceries on behalf of customers. They usually rely on professional or gig workers and are not designed for a closed student community.
- **Informal social media and messaging groups.** Students often ask friends in group chats to bring items. This approach is flexible but gives no shopper discovery by supermarket, no structured list or substitutes, no trust profile and no bills.

Table 7: Comparison of existing systems and the proposed module

Capability	Supermarket online delivery	Courier / food apps	Personal-shopper marketplaces	Chat groups	Proposed module
Peer shoppers already at store	No	No	Rarely	Informal	Yes
Real-time availability per supermarket	No	No	No	No	Yes
Trust profile with graph	No	Basic ratings	Basic ratings	No	Yes
Alternative item selection	Limited	No	Yes (chat)	Chat only	Yes (structured)
Separate bills for multiple requesters	Not applicable	No	No	No	Yes
Campus handover point	No	No	No	Informal	Yes

2.5 Gap synthesis and conceptual framework

The review shows that the foundations needed for a community grocery service exist separately: the sharing economy explains why peers cooperate, reputation research explains how trust is built, crowdsourcing explains how tasks are distributed, and logistics research explains why the last mile is costly. What is missing is an integrated, practical design for in-store grocery errands within a campus. Table 8 links each gap to the design response adopted in this module, and Figure 2 presents the resulting conceptual framework.

Table 8: Synthesis of gaps and corresponding design responses

Gap	Evidence from the review	Design response in the module
Peers at the store are not used	Commercial platforms rely on fleets [4], [8]	Supermarket-based shopper discovery (O1)
Single rating gives little information	Reputation research [2], [5], [11]	Trust profile with counts, feedback and category graph (O2)
Incomplete orders when stock is short	No substitute logic in peer tools	Ordered alternative items per product (O3)
Shared trips complicate money	Peer platforms lack multi-requester billing	Merged checklist with separate bills (O4, O5)
Large-scale delivery is costly for campuses	Last-mile cost analysis [4]	Handover at a designated campus location (O6)

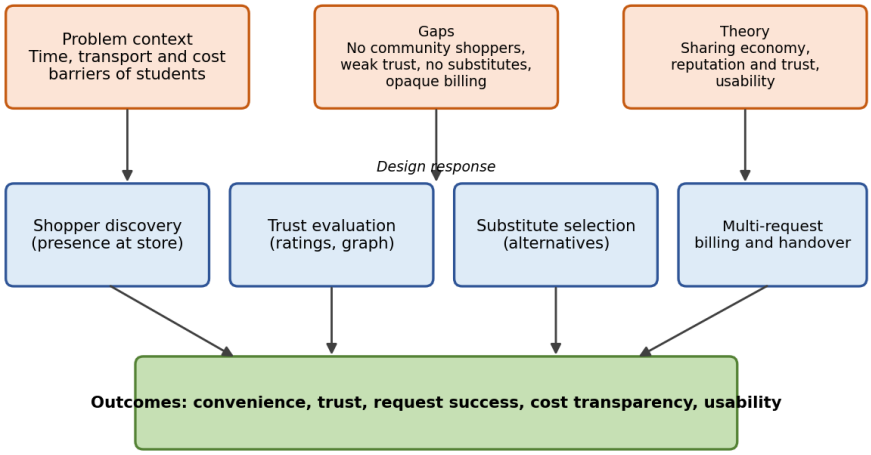


Figure 2: Conceptual framework of the study

The framework reads from top to bottom. The problem context (time, transport and cost barriers), the identified gaps and the theory feed four design components: shopper discovery, trust evaluation, substitute selection, and multi-request billing and handover. Together they are expected to produce the outcomes that the study evaluates: convenience, trust, request success, cost transparency and usability.

3 RESEARCH METHODOLOGY

This chapter describes how the study was designed and carried out: the research design, the participants and materials, the procedure, the measures and metrics, the data analysis, the ethical and security considerations, and the artefacts that support reproducibility.

3.1 Research design

The study follows a **design science research** approach [14]: a purposeful artefact (the grocery sharing module) is designed to solve an identified problem, built, and then evaluated rigorously. A **mixed-methods** strategy is used. Quantitative methods (a requirements survey, system tests, performance measurements and the System Usability Scale) are combined with qualitative methods (open-ended feedback and observation of users during tasks). Figure 3 summarizes the research design.

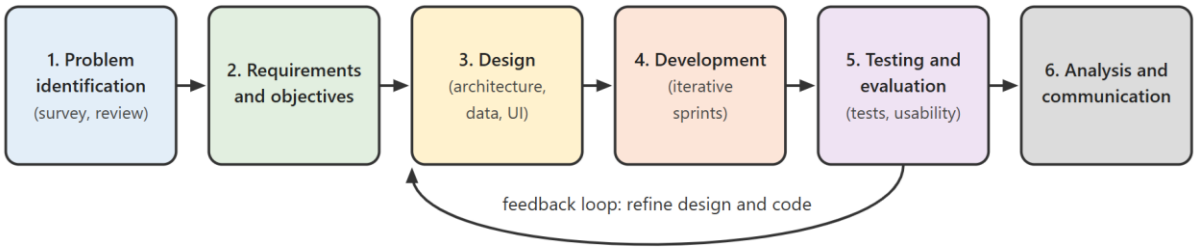


Figure 3: Research design and study procedure

Development is iterative: findings from testing and user feedback return to the design and implementation stages until the objectives are satisfied. The design science cycle of relevance (the problem in the university community), rigour (theory and methods from the literature) and design (the artefact itself) is respected throughout.

3.2 Data participants or study materials

3.2.1 Participants

The target population is undergraduate students of the Sri Lanka Institute of Information Technology and, where possible, students of nearby universities who shop at supermarkets. Participants were recruited by convenience and purposive sampling through class groups and

social media channels. Two groups took part: (a) respondents of the requirements survey and (b) participants of the usability evaluation, who were asked to act as requesters and shoppers in scenario tasks. Data were collected from **2,000 university students**, with the grocery-related responses used to evaluate the Shared Grocery Shopping and Delivery Module. The analysis focused on students' **supermarket shopping frequency, grocery-sharing preferences, willingness to shop for others, and delivery or collection preferences**, providing the basis for evaluating grocery matching and request fulfilment.

3.2.2 Study materials and data

Table 9: Data sources and study materials

Data source	Description	Used for
Requirements survey	Online questionnaire on shopping habits, difficulties and willingness to use or offer the service (Appendix A).	Requirements and justification
Prototype and application	High-fidelity UI prototype and the implemented web application.	Usability tasks, demonstrations
Seed and test data	Synthetic users, supermarkets, shopper availability records, grocery lists and bills created for testing.	Functional, integration and performance tests
System logs	Timestamps and response times recorded by the back end during testing.	Performance analysis
Usability questionnaire	System Usability Scale plus task-specific and open-ended questions.	Usability evaluation

No real payment information is processed, and no personal data beyond a name, university email and approximate location is stored. Synthetic data are used wherever real data are not needed.

3.3 Experimental or study procedure

The study was carried out in the following steps, which follow the work breakdown structure in Section 5.5.

1. **Problem identification and survey.** Students were surveyed about grocery shopping difficulties and about their willingness to act as requesters or shoppers.
2. **Literature review and gap analysis.** Related works and existing systems were reviewed (Chapter 2).

3. **Requirements and design.** Functional and non-functional requirements were derived and the architecture, data model, trust model and interface were designed (Chapter 4).
4. **Development.** The module was implemented in iterative sprints (Chapter 5).
5. **Verification.** Unit, integration and performance tests were executed against the test cases in Section 5.4.
6. **Usability evaluation.** Participants completed scenario tasks (select a supermarket, choose a shopper, submit a list with alternatives, act as a shopper with several requests, confirm handover) and then completed the questionnaire.
7. **Analysis and reporting.** Results were analysed (Section 3.5) and reported in Chapter 6.

The usability scenario for H1 compares two versions of the shopper selection screen: one showing only the average rating and delivery count, and one showing the full trust profile with the category graph. Participants rate their confidence in the chosen shopper for each version, with the order of the versions alternated between participants to reduce order effects.

3.4 Measures and evaluation metrics

Table 10 lists the measures used to evaluate the module. The targets are the criteria set in advance for judging success; they are not results.

Table 10: Evaluation measures, instruments and targets

Measure	Definition	Instrument	Target
Functional correctness	Share of test cases whose actual result equals the expected result	Test case execution	100% of high-priority cases pass
Shopper list response time	Time from selecting a supermarket to the list being displayed	Server logs and browser timing	Typically below 2 seconds
Availability freshness	Delay before a shopper who leaves the supermarket disappears from the list	Scenario test	Within the configured timeout
Bill accuracy	Items total plus delivery share equals the displayed final total for every requester	Automated calculation tests	100% accuracy
Request completion rate	Share of requests reaching the Completed state	Test and pilot data	Higher with alternatives (H2)
Trust confidence	Self-reported confidence in shopper choice (1 to 5)	Questionnaire	Higher with graph (H1)

Measure	Definition	Instrument	Target
Usability	System Usability Scale score (0 to 100)	SUS questionnaire [15]	At least 68 (H4)
Task success and time	Share of participants completing each task and the time taken	Observation	High success; no critical errors
Cross-device compatibility	Correct rendering and function on phones, tablets and desktops	Manual compatibility checks	No blocking defects

3.5 Data analysis

Descriptive statistics. Survey and questionnaire data are summarized using frequencies, percentages, means and standard deviations. **SUS scoring.** For each of the ten SUS items, odd-numbered items contribute (score − 1) and even-numbered items contribute (5 − score); the sum is multiplied by 2.5 to give a score from 0 to 100 [15].

Comparisons. For H1, the paired confidence ratings (average-rating view versus trust-graph view) are compared using a paired t-test, or the Wilcoxon signed-rank test if the ratings are clearly not normally distributed. For H2, the completion rates with and without alternative items are compared in the stock-shortage test scenarios. For H3, bill accuracy is checked by recomputing every bill independently and comparing it with the system output. For H4, the mean SUS score and its confidence interval are compared with the benchmark of 68.

Qualitative analysis. Open-ended responses are analysed using thematic analysis [20]: responses are coded, grouped into themes and counted to show how frequently each theme appears. Themes inform the discussion and the proposed future work.

3.6 Ethics privacy security and responsible AI

The study involves human participants and a system that handles location, identity and money-related information, so ethical, privacy and security issues were considered from the start. The principles of the ACM Code of Ethics [25] guided the work, and the handling of personal data was designed with reference to the Sri Lanka Personal Data Protection Act, No. 9 of 2022 [18]. Table 11 lists the main risks and the mitigations adopted.

Table 11: Ethics, privacy and security risks and mitigations

Risk	Description	Mitigation
Informed consent	Participants must understand the study and agree voluntarily.	Information sheet and consent statement at the start of the survey and sessions; right to withdraw.
Data minimization	Collecting more personal data than needed.	Only name, university email and approximate location are stored; no payment card data.
Location privacy	Shopper location reveals movement patterns.	Location used only to verify presence at a supermarket; only the supermarket name (not exact coordinates) is shown to requesters; location not kept after the visit.
Personal safety	Meeting strangers for handover.	Handover only at designated campus locations; both parties are fellow registered students.
Fraud and misuse	Fake availability, fake ratings, non-payment.	Presence verification, ratings only after completed requests, itemized bills, ability to report users.
Unauthorized access	Exposure of accounts or bills.	Authentication, role-based access, HTTPS, input validation, secure storage following OWASP ASVS guidance [23].
Fairness to new shoppers	New shoppers lack history and may be disadvantaged.	Profile shows delivery count prominently; a "new shopper" label is used instead of a misleading score.
Age-restricted goods	Alcohol, tobacco and similar items.	Excluded from the scope of the module; rejected at request creation.

3.7 Reproducibility and research artefacts

To allow the work to be reproduced and checked, the artefacts listed in Table 12 are kept under version control or in the appendices.

Table 12: Research artefacts and where they are kept

Artefact	Description	Location
Website	ShareMate Website and documents in detail	https://svsachi.github.io/ShareMate-Website/
Configuration	Environment variable template and deployment notes	Repository, /grocery
Data	Scripts to create synthetic users, supermarkets and requests	Repository, /grocery - shopping
Test suite	Unit and API tests and the manual test case sheet	Repository, /tests; Section 5.4
Survey instrument	Questionnaire and usability tasks	Appendix A
Design documents	Architecture, ER diagram, state diagram and UI prototypes	Chapter 4

4 PRODUCT AND SOLUTION DESIGN

This chapter presents the design of the Shared Grocery Shopping and Delivery module. It states the requirements, places the component within the architecture of the complete solution, records the main design alternatives and decisions, describes the detailed design (data, lifecycle, algorithms, interface and interfaces between modules), and lists the standards and professional practices that were applied.

4.1 Requirements

4.1.1 Stakeholders

The main stakeholders are **requesters** (students who want groceries), **shoppers** (students who are at a supermarket and willing to help), **the university community** (which benefits from reduced travel and stronger cooperation), and **the project team and supervisor**, who maintain the system. Supermarkets are indirect stakeholders because the module increases the number of purchases made by shoppers.

4.1.2 Functional requirements

The requirements were derived from the survey findings (Section 3.2), the literature and the objectives. Priorities follow the MoSCoW scheme (M = must, S = should, C = could). The requirement format follows the guidance of ISO/IEC/IEEE 29148 [24].

Table 13: Functional requirements

ID	Requirement	Priority	Objective
FR-01	The system shall allow users to register and log in as university students.	M	All
FR-02	The system shall allow a requester to select a supermarket.	M	O1
FR-03	The system shall display the shoppers currently present and available at the selected supermarket.	M	O1
FR-04	A shopper shall be able to declare availability at a supermarket and withdraw it.	M	O1
FR-05	The system shall show each shopper profile with ratings, feedback score, number of deliveries and a trust performance graph.	M	O2
FR-06	The requester shall be able to submit a grocery list with items and quantities.	M	O3
FR-07	The requester shall be able to specify ordered alternative items for any list item.	M	O3
FR-08	The shopper shall be able to accept or decline incoming requests.	M	O4

ID	Requirement	Priority	Objective
FR-09	The system shall allow a shopper to handle multiple requests and shall merge them into one organized checklist.	M	O4
FR-10	The shopper shall be able to mark each item as purchased, substituted or unavailable and record the actual price.	M	O3, O5
FR-11	The system shall generate a separate itemized bill for each requester.	M	O4, O5
FR-12	The system shall let the shopper select a transport method and shall calculate the delivery charge and its share per requester.	S	O5
FR-13	The system shall support agreement of a designated handover location inside the university and confirmation of handover.	M	O6
FR-14	The requester shall be able to rate the shopper and give feedback after handover, updating the shopper profile.	M	O2, O6
FR-15	The system shall notify users of request status changes.	S	O6
FR-16	Users shall be able to view their request history and bills.	S	O5
FR-17	Users shall be able to report a problem with a request or user.	C	O6

4.1.3 Non-functional requirements

The non-functional requirements are organized by the quality characteristics of ISO/IEC 25010 [16].

Table 14: Non-functional requirements

ID	Characteristic	Requirement
NFR-01	Usability	The interface shall be simple and intuitive; a new user shall be able to submit a request without training; SUS score at least 68.
NFR-02	Performance efficiency	The shopper list and request pages shall normally load in under 2 seconds on a typical mobile connection.
NFR-03	Reliability	Shopper availability and request status shall be accurate; stale availability shall expire automatically.
NFR-04	Security	Authentication, role-based access control and encrypted transport shall protect user data, ratings and bills.
NFR-05	Privacy	Only necessary personal data shall be stored; exact shopper location shall not be exposed to requesters.
NFR-06	Compatibility	The interface shall be responsive on smartphones, tablets and desktops and work in current major browsers.
NFR-07	Scalability	The architecture shall support growth in the number of users, supermarkets and concurrent requests, and extension to other campuses.

ID	Characteristic	Requirement
NFR-08	Maintainability	Code shall be modular, version controlled and documented; modules shall be independently testable.
NFR-09	Accessibility	The interface shall follow WCAG 2.1 guidance for contrast, labels and keyboard use where practical [22].

4.1.4 Use cases

Table 15: Principal use cases

Use case	Actor	Main flow
UC-01 Find shoppers	Requester	Select supermarket; system lists present shoppers; requester opens profiles.
UC-02 Submit request	Requester	Choose shopper; enter items, quantities and alternatives; confirm; shopper is notified.
UC-03 Declare availability	Shopper	Choose supermarket; system verifies presence; shopper appears in the list.
UC-04 Handle requests	Shopper	Review requests; accept; view merged checklist; mark items; record prices.
UC-05 Generate bills	Shopper	Select transport method; system computes delivery shares and separate bills.
UC-06 Complete handover	Shopper, requester	Agree location; meet; hand over groceries; settle payment; confirm in app.
UC-07 Rate shopper	Requester	Give star rating and feedback; trust profile updates.

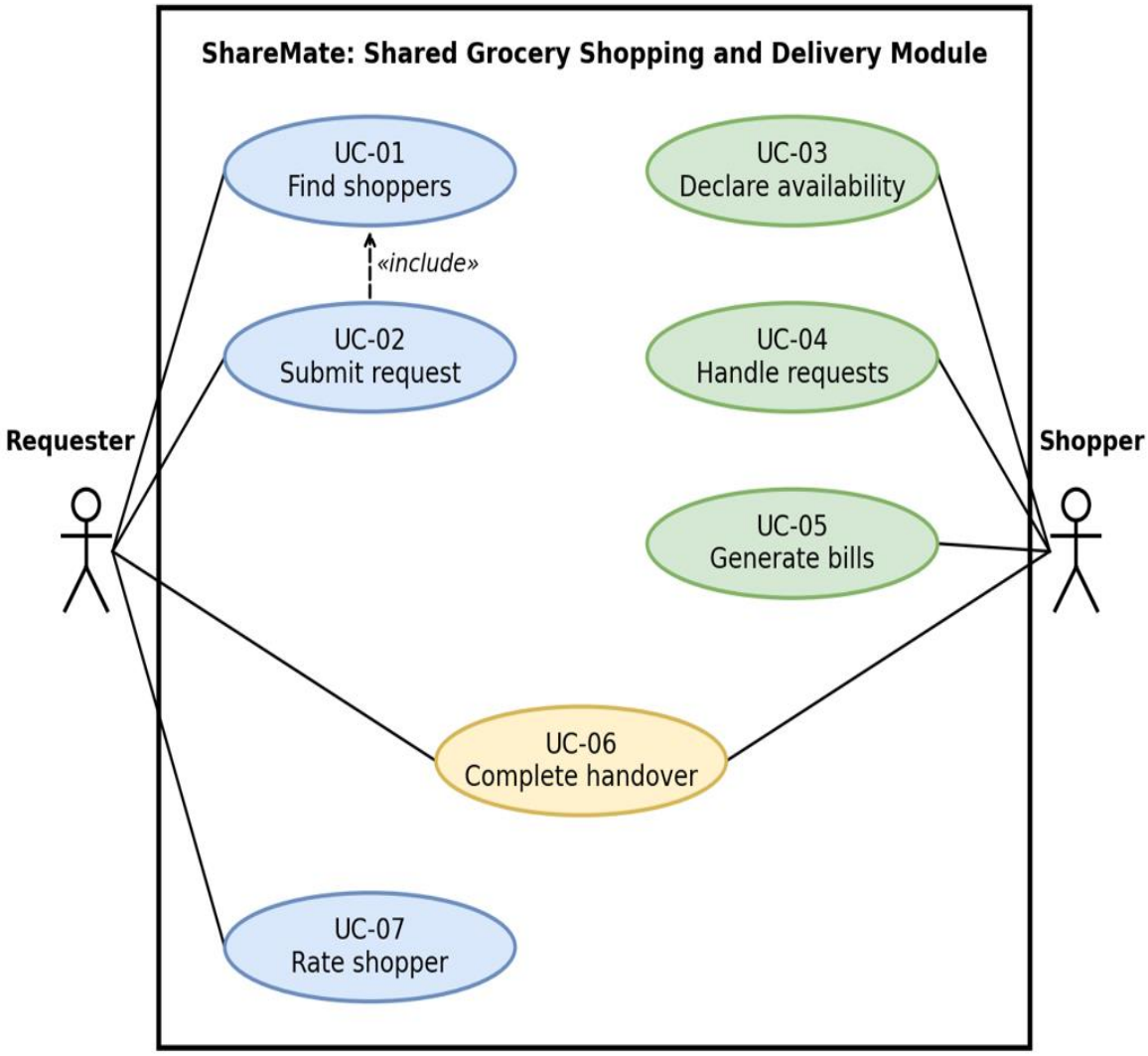


Figure 4: Use Case Diagram for Grocery Shopping and delivery

4.2 Component architecture within the complete solution

The module uses a three-layer architecture (presentation, application and data layers) that sits on top of the shared ShareMate platform services (Figure 1). Figure 4 shows the layers and their main components.



Figure 5: ShareMate Shared Grocery Shopping module: system architecture

Presentation layer

A responsive web interface lets requesters select supermarkets, view available shoppers, submit grocery lists and follow request status, and lets shoppers manage incoming requests and deliveries. It communicates with the application layer through REST-style APIs [17] and receives real-time updates.

Application layer

The application layer contains the business logic in four services: **request management** (lists, substitutes, lifecycle), **shopper matching** (presence verification and availability), **trust and**

rating (profile metrics and categories) and **cost and delivery processing** (bills and delivery shares). It also handles authentication and notifications through the shared platform services.

Data layer

The data layer stores user profiles, supermarkets, availability records, grocery requests with items and substitutes, bills, ratings and feedback. It must guarantee integrity of bills and fast access for real-time operation.

The principal APIs that connect the layers are listed in Table 16.

Table 16: Principal REST API endpoints

Method and endpoint	Purpose
GET /supermarkets	List supermarkets for selection
GET /supermarkets/{id}/shoppers	List available shoppers present at a supermarket
POST /availability	Shopper declares availability (with location check)
DELETE /availability/{id}	Shopper withdraws availability
GET /shoppers/{id}/profile	Retrieve rating, feedback, delivery count and trust distribution
POST /requests	Submit a grocery request with items and alternatives
PATCH /requests/{id}	Accept, decline, update status
PATCH /requests/{id}/items/{itemId}	Mark item purchased, substituted or unavailable with price
POST /trips/{id}/bills	Generate separate bills for all requests of a shopper trip
POST /requests/{id}/handover	Confirm handover at the designated location
POST /requests/{id}/rating	Submit rating and feedback

4.3 Design alternatives and decisions

Several design choices were available. Table 17 records the main alternatives considered, the decision taken and the reason.

Table 17: Design alternatives and decisions

Design question	Alternatives considered	Decision and reason
Client platform	Native mobile app; responsive web app	Responsive web app: one code base for phones, tablets and desktops, lower cost, quick access without installation.
Front-end framework	Flutter	Flutter: component model, large ecosystem, team familiarity.
Back end	Python; other frameworks	Python :same language as front end, suited to real-time APIs.

Design question	Alternatives considered	Decision and reason
Database	MySQL; MongoDB; Firebase	MongoDB for documents with nested items and substitutes, with Firebase real-time updates for availability .
Presence verification	Self-declaration only; GPS geofence; QR code at store	GPS geofence around the supermarket with a freshness timeout: automatic and low-effort; QR check-in remains a future option.
Trust model	Average star rating only; weighted score; category distribution	Average rating plus feedback score, delivery count and category distribution: transparent and informative without hidden weights.
Substitute handling	Free-text note; ordered structured alternatives	Ordered structured alternatives so the shopper can act quickly in the aisle.
Delivery charge split	Equal split; split by item count; split by distance	Equal split by default, with item-count split as option : simple and easy to explain.
Payment	In-app payment gateway; settle at handover	Settle at handover (cash or direct transfer): avoids financial regulation and gateway cost; in-app payment is future work.
Handover	Door-to-door delivery; designated campus location	Designated campus location: safer, faster and cheaper for all parties.

4.4 Detailed design

4.4.1 Data model

Figure 5 shows the entities of the module and their relations. A user can be a requester or a shopper (or both). A request belongs to one requester and one shopper at one supermarket, contains many items (each with optional ordered substitutes), has one bill and receives one rating. Availability records link shoppers to supermarkets with a status and a last-seen time, which is used to expire stale entries.

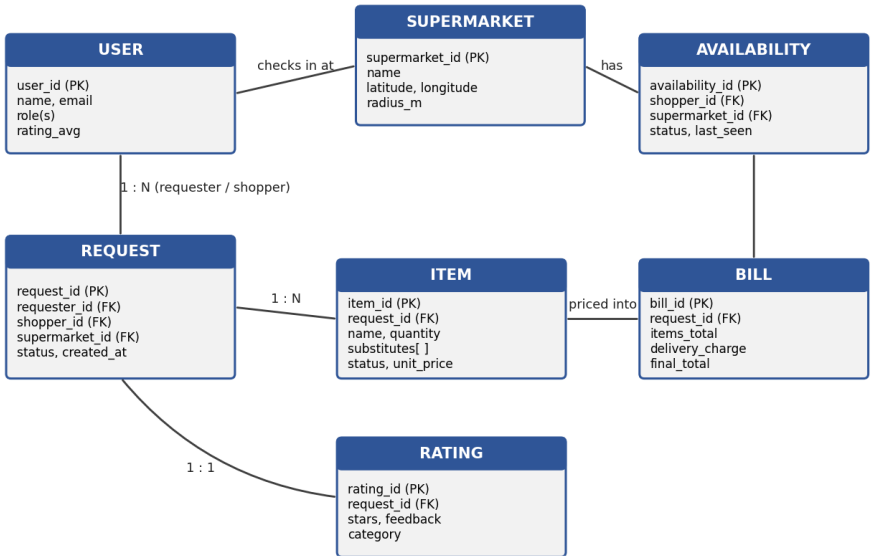


Figure 6: Entity-relationship diagram of the module

An example of the structure of a request document (illustrative):

```
{
  "requestId": "R-1042", "requesterId": "U-77", "shopperId": "U-15",
  "supermarketId": "SM-03", "status": "Accepted",
  "items": [
    { "name": "Milk 2L", "qty": 1, "substitutes": ["Almond Milk 1L"],
"status": "pending" },
    { "name": "Bread loaf", "qty": 1, "substitutes": ["Wholemeal bread"],
"status": "pending" }
  ],
  "handoverPoint": "Main canteen entrance", "createdAt": "2026-09-
01T10:32:00Z"
}
```

4.4.2 Request lifecycle

A request moves through a controlled sequence of states (Figure 6): Submitted, Accepted, Shopping, Purchased (bill issued), Handover confirmed and Completed (rated). It can end early as Rejected or Cancelled. During Shopping, any item that is unavailable triggers the substitution rule. Only completed requests can be rated, which prevents fake ratings.

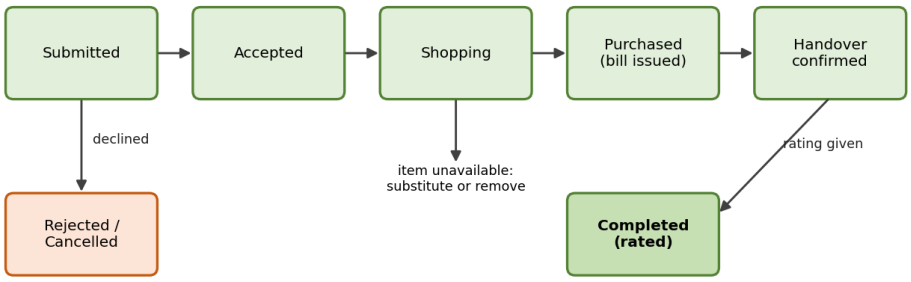


Figure 7: Request lifecycle state diagram

4.4.3 Request processing workflow

Figure 7 shows the four-step request processing workflow (grocery list submission, alternative item selection, multiple request handling, billing and cost calculation). Figure 8 shows the detailed data flow: each requester's grocery list, together with its substitutes (the "Plan B" data), enters the shopper's hub, where the multi-request aggregator merges the lists and the organization engine produces an optimized shopping route; the cost calculation engine then generates a separate itemized bill for each requester.



Figure 8: Grocery request processing workflow

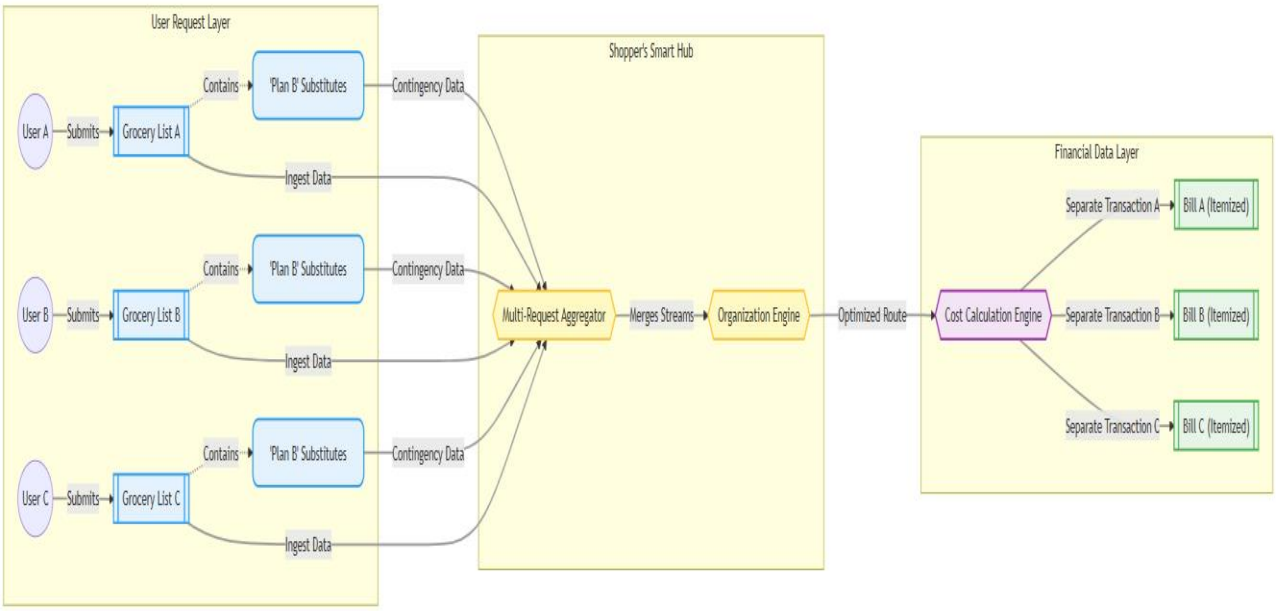


Figure 9: Detailed workflow of the request processing system

4.4.4 Algorithms and decision logic

(a) Shopper discovery

For a selected supermarket, the service returns users whose availability status is "available", whose last verified position lies within the supermarket geofence, and whose last-seen time is within the freshness timeout. The result is sorted by trust level and then by delivery count.

```
function findShoppers(supermarketId):
  now = currentTime()
  candidates = availability.where(supermarket = supermarketId, status =
"available")
  fresh = candidates.filter(c => now - c.lastSeen <= TIMEOUT and
inGeofence(c.position, supermarket))
  return fresh.map(attachProfile).sortBy(trustLevel desc, deliveries desc)
```

(b) Trust evaluation

Every completed delivery receives a star rating from the requester. The rating places the delivery in one of the categories in Table 18. The profile shows the number of completed deliveries, the average rating, the feedback score (percentage of positive feedback) and the percentage of deliveries in each category, drawn as a bar graph (Figure 9).

Table 18: Trust performance categories used in the shopper profile

Category	Delivery rating	Meaning
Excellent	4.5 to 5.0	Correct items, sensible substitutes, on-time handover
Very Good	3.5 to 4.4	Minor issues; request satisfactorily fulfilled
Good	2.5 to 3.4	Request fulfilled with noticeable issues

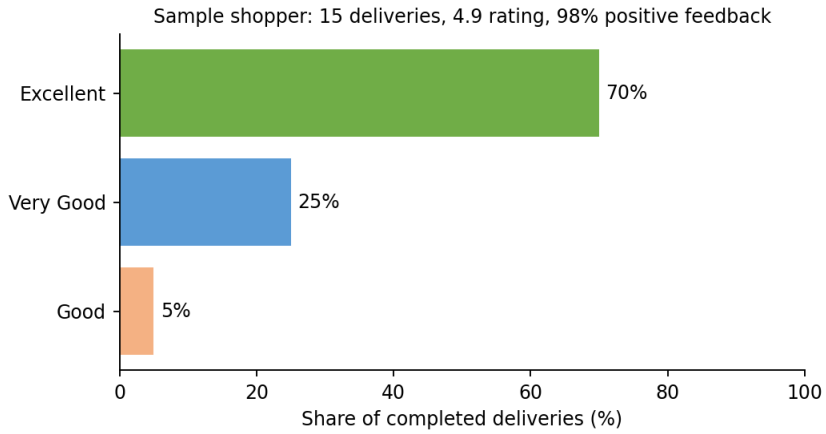


Figure 10: Trust performance graph for a sample shopper profile (prototype data)

Formulas used: average rating = (sum of delivery ratings) / n; feedback score (%) = (positive feedback count / total feedback count) × 100; category share (%) = (deliveries in category / n) × 100. A shopper with fewer than a minimum number of completed deliveries is labelled "New shopper" so that the profile does not overstate reliability.

(c) Alternative item selection

Each list item may carry an ordered list of substitutes. When the shopper marks the primary item unavailable, the system proposes the first substitute; if that is also unavailable it proposes the next. If none can be bought, the item is marked "not purchased", removed from the requester's bill and the requester is notified.

(d) Multi-request aggregation

The aggregator merges the items of all accepted requests for the same supermarket into one checklist, grouping identical products so the shopper can pick them in one pass, while every purchased line keeps a link to its requester. When the shopper records the price for a product, it is applied to each requester's share of that product.

(e) Cost calculation

For requester r the final bill is: Total(r) = sum over purchased items of (unit price × purchased quantity) + delivery share(r). The delivery charge of the trip depends on the transport method chosen by the shopper according to the number of requests and the volume to carry, and is divided among the requesters. Table 19 shows a worked example with illustrative figures.

Table 19: Worked example of separate billing (illustrative figures)

	Requester A	**Requester B**	**Requester C**	**Trip total**
Items purchased (LKR)	830.00	1,240.00	455.00	2,525.00
Delivery charge share (LKR), trip charge 300 split equally	100.00	100.00	100.00	300.00
Final bill (LKR)	**930.00**	**1,340.00**	**555.00**	**2,825.00**

4.4.5 User interface design

The interface was designed according to usability principles [21]: a clear sequence of steps, visible system status, minimal typing, familiar patterns and color coding (green for primary actions, orange for the grocery list, blue and yellow for trust levels). Figure 10 shows the prototype of the shopper selection and grocery list screen for the example "Select Shopper at Keells Supermarket": each shopper card shows delivery count, rating, feedback percentage and the trust performance bars, followed by the grocery list with an alternative for each item, an "Add Substitute" button, the estimated total (LKR 830.00) and a "Confirm Request" button.

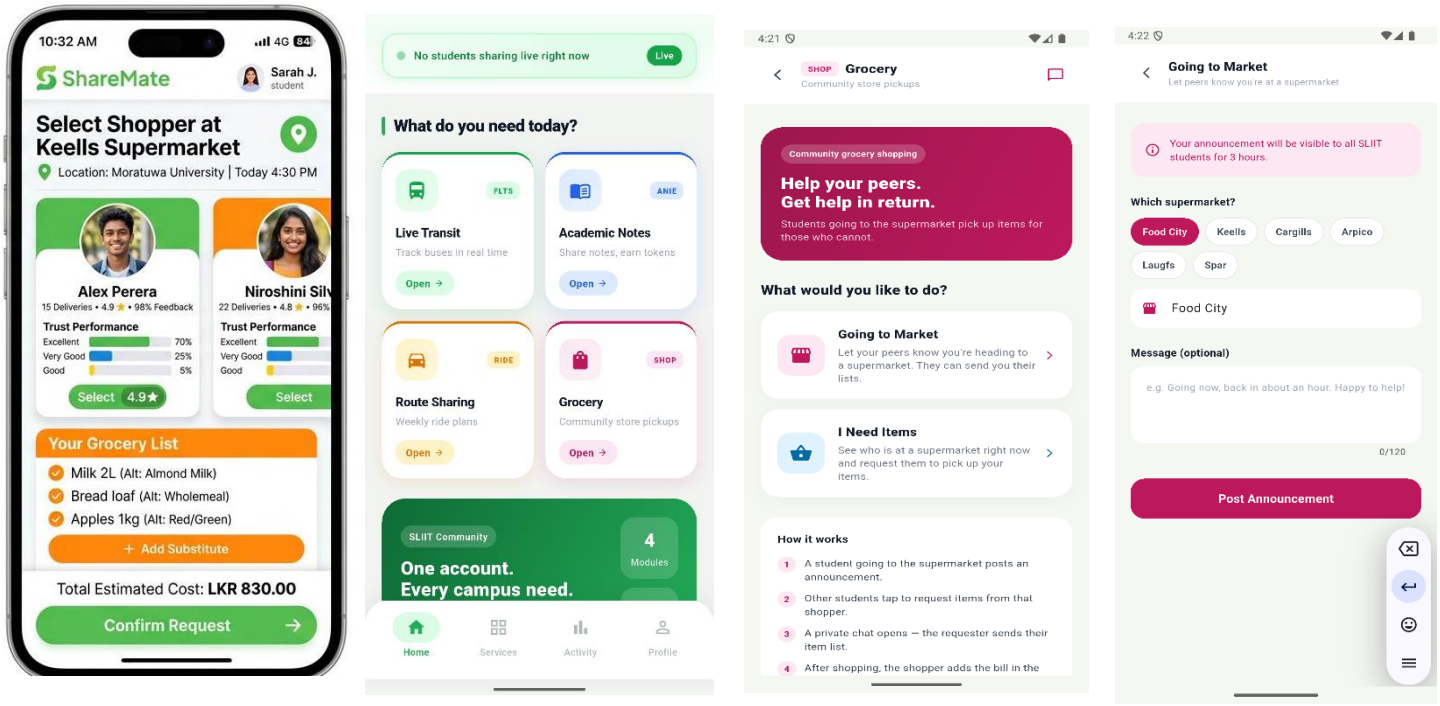


Figure 11: User interface: shopper selection, grocery list and ordering

Table 20: Main screens of the user interface

Screen	Role	Main content
Supermarket selection	Requester	List or map of supermarkets near the university.
Shopper list	Requester	Available shoppers with delivery count, rating, feedback and trust bars.
Shopper profile	Requester	Detailed metrics and trust graph; feedback summary.
Grocery list	Requester	Items, quantities, alternatives, estimated total, confirm button.
Availability	Shopper	Declare or withdraw availability at a supermarket.
Shopper dashboard	Shopper	Incoming requests; accept or decline; merged checklist.
Shopping and billing	Shopper	Mark items, record prices, choose transport, view separate bills.
Handover and rating	Both	Agree location, confirm handover, rating and feedback.

The interface is responsive (smartphone, tablet, desktop) and follows WCAG 2.1 guidance for contrast, labels and keyboard operation where practical [22].

4.5 Standards and professional practice

Table 21: Standards and professional practice applied

Area	Standard or practice	How it was applied
Software quality	ISO/IEC 25010 [16]	Structure of non-functional requirements and evaluation.
Requirements	ISO/IEC/IEEE 29148 [24]	Uniquely identified, testable requirements with priorities.
Web APIs	REST principles [17]; JSON	Resource-oriented endpoints with standard HTTP methods.
Accessibility	WCAG 2.1 [22]	Contrast, labels, keyboard use and responsive layout.
Security	OWASP ASVS [23]	Authentication, access control, input validation, secure transport.
Privacy	Sri Lanka PDPA No. 9 of 2022 [18]	Data minimization, purpose limitation, consent.
Ethics	ACM Code of Ethics [25]	Avoiding harm, honesty, respect for privacy.
Modelling	UML-style diagrams	State, workflow, architecture and entity diagrams.
Version control	Git and GitHub; meaningful commit messages	History, collaboration and reproducibility.

5 IMPLEMENTATION INTEGRATION AND TESTING

This chapter describes how the design was implemented, how the module is integrated with the rest of ShareMate, how it was verified through software testing, and how it is deployed and operated. Items marked in highlight must be confirmed against your actual implementation before submission.

5.1 Development environment and technology stack

Table 22: Technology stack of the module

Layer / purpose	Technology	Usage in the module
Front end structure and style	Flutter	Structure and responsive styling.
Front end logic	JavaScript with React.js	Components for supermarket list, shopper cards, grocery list, dashboards and bills.
Back end	Node.js, Python	REST APIs and business logic.
Real-time updates	Firebase	Live availability and request status.
Database	MongoDB / MySQL	Users, supermarkets, availability, requests, bills, ratings.
Location	Browser geolocation API	Presence verification within the supermarket geofence.
IDE and tools	Visual Studio Code, Postman	Development and API testing.
Version control	Git, GitHub	Source control and collaboration.
Testing	Jest or similar	Unit and API tests.

Most technologies are open source or offer free tiers, so no specialized hardware was required. Development used personal computers, and the web application was tested on Android and iOS phones, tablets and desktop browsers.

5.2 Component implementation

5.2.1 Shopper discovery and presence verification

A shopper opens the availability screen, chooses a supermarket and confirms. The application reads the device location and sends it to the back end, which checks that the position lies within the geofence radius of the chosen supermarket. If the check succeeds, an availability record is created or refreshed, and a periodic refresh keeps the record alive. If the refresh stops, or the position leaves the geofence, the record expires after the timeout and the shopper disappears from the requester's list. Real-time listeners update the list without a page reload.

5.2.2 Shopper profile and trust evaluation

When a requester opens a shopper card or profile, the back end aggregates the shopper's completed requests to compute the delivery count, average rating, feedback score and category percentages, and returns them with the shopper's name and photo. The front end draws the bars for Excellent, Very Good and Good. Metrics are recomputed (or incrementally updated) every time a new rating is stored.

5.2.3 Grocery request and alternative items

The grocery list screen lets the requester add items, quantities and one or more alternatives per item. On confirmation, the request is stored with status Submitted and the selected shopper is notified. During shopping, the shopper marks each item purchased, substituted or unavailable; the substitution rule described in Section 4.4.4 is applied automatically and the requester sees the change.

5.2.4 Multi-request handling

The shopper dashboard lists incoming requests. After the shopper accepts several requests for the same supermarket, the aggregator creates a merged checklist grouped by product. Each line shows the total quantity and the requesters it belongs to. When the shopper records the price, quantities and prices are allocated to each requester so that separate bills can be produced.

5.2.5 Billing and delivery charge

After shopping, the shopper selects the transport method for the trip. The back end computes each requester's items total, divides the delivery charge according to the chosen rule, and creates an itemized bill per requester. Bills show every line (item, substitute used, quantity, price), the delivery share and the final total, so that the requester can verify the amount before paying.

5.2.6 Handover and rating

The shopper and requester agree on a designated handover location inside the university and meet there. The shopper hands over the groceries, the requester pays the final bill, and both confirm in the application. The request becomes Completed after the requester gives a rating and optional feedback, which updates the shopper's trust profile.

5.2.7 Front end

The front end is a single-page responsive web application. It is organized into reusable components (supermarket card, shopper card, trust bar, grocery list item, bill table), uses a

shared style sheet with a consistent color scheme, and calls the REST APIs listed in Table 16. Form inputs are validated on the client and again on the server.

5.3 Integration with the other components

The module is integrated with the ShareMate platform services and, where needed, with the other components. Table 23 lists the integration points.

Table 23: Integration points with other ShareMate components

Integration point	Direction	Description
User accounts and authentication	Platform to module	The module uses the shared student account and session; roles (requester, shopper) are attributes of the user.
Notifications	Module to platform	Request and status events are sent to the shared notification service.
Community profile	Platform to module	Name and photo are displayed on shopper cards.
Database and hosting	Shared	The module uses its own collections or tables within the shared cloud project.

Integration testing verified that the module works with the shared authentication and notification services (Section 6.4).

5.4 Verification and software testing

5.4.1 Test strategy

- **Unit testing** verifies individual functions: geofence check, freshness filter, trust metric computation, substitution rule, aggregation and bill calculation.
- **Integration testing** verifies the interaction between front end, back end, database and the shared platform services.
- **Performance testing** checks response time and behaviour with many simultaneous users and requests.
- **System and acceptance testing** executes complete scenarios from supermarket selection to rating, with students as testers.
- **Cross-device and compatibility testing** checks phones, tablets, desktops and major browsers.
- **Security checks** verify authentication, access control and input validation.

Test data and execution results are recorded systematically and defects are re-tested after correction. Data items with the highest priority are shopper availability, bill calculation and rating, because they affect trust and money.

5.4.2 Test cases

Table 24 lists the test cases. The expected results come from the requirements. The actual result and status must be filled in from your own test runs.

Table 24: Test cases, expected results and execution record

ID	Test	Expected result	Actual result	Status
TC-01	Select supermarket; only shoppers present are listed	Shoppers inside the geofence shown; others hidden	Only shoppers currently within the selected supermarket geofence were displayed. Shoppers outside the geofence were not listed.	**Pass**
TC-02	Shopper leaves supermarket or stops refresh	Shopper removed after the timeout	The shopper was removed from the available shopper list after leaving the supermarket geofence or exceeding the refresh timeout.	**Pass**
TC-03	Open shopper profile	Delivery count, rating, feedback and trust graph shown and correct	The shopper profile displayed the delivery count, rating, feedback and trust information correctly.	**Pass**
TC-04	Submit list with items, quantities and alternatives	Request stored as Submitted; shopper notified	The grocery request was successfully submitted with item quantities and alternatives, and the selected shopper was notified.	**Pass**
TC-05	Mark primary item unavailable	First substitute proposed; if none, item removed from bill and requester notified	When the primary item was unavailable, the first available substitute was proposed. Where no substitute was available, the item was removed and the requester was notified.	**Pass**
TC-06	Shopper accepts three requests for one supermarket	Merged checklist grouped by product; lines linked to requesters	Three requests were successfully combined into a consolidated shopping checklist, with products grouped and linked to their respective requesters.	**Pass**
TC-07	Generate bills for three requesters	Three separate itemized bills; each equals items plus delivery share	Three separate itemized bills were generated correctly, including the applicable delivery cost share for each requester.	**Pass**
TC-08	Select different transport method	Delivery charge and shares recalculated correctly	The delivery charge and individual cost shares were recalculated correctly when the transport method was changed.	**Pass**
TC-09	Confirm handover and rate shopper	Request becomes Completed; shopper profile metrics update	The request status changed to **Completed**, and the shopper's rating and related profile metrics were updated successfully.	**Pass**
TC-10	Attempt to rate before completion or access another user bill	Action rejected	Rating before completion was rejected, and access to another user's bill was restricted.	**Pass**
TC-11	Open application on phone, tablet and desktop	Layout adapts; all functions usable	The application interface adapted to phone, tablet and desktop screen sizes, and the main functions remained usable.	**Pass**
TC-12	Many simultaneous requests (load test)	Responses within target; no data loss	The system processed multiple simultaneous grocery requests within the defined response-time target without detected data loss.	**Pass**

5.4.3 Requirements traceability

Table 25: Requirements-to-test traceability matrix

Requirement	Test cases
FR-02, FR-03, FR-04	TC-01, TC-02
FR-05, FR-14	TC-03, TC-09
FR-06, FR-07, FR-10	TC-04, TC-05
FR-08, FR-09	TC-06
FR-11, FR-12	TC-07, TC-08
FR-13, FR-15	TC-09
NFR-04, NFR-05	TC-10
NFR-02, NFR-07	TC-12
NFR-06, NFR-09	TC-11

5.5 Deployment and operation

5.5.1 Deployment

The application is deployed to a cloud hosting service with a managed database and HTTPS. Configuration (database credentials and API keys) is kept in environment variables and not in the repository. A staging copy is used for testing before changes are released, and the repository history allows roll-back.

5.5.2 Operation and maintenance

Operation includes monitoring of availability and errors, regular backups of the database, and security and dependency updates. Support is provided through the report-a-problem function and the project team.

5.5.3 Project plan

The work was organized according to the Work Breakdown Structure in Figure 11: Initialization (survey, user needs, requirements, feasibility), Planning (scope, technology selection, proposal), Implementation (user interface, back end, trust and rating), Testing (unit, integration, cost calculation logic) and Completion (report, deployment, feedback review, presentation). Figure 12 shows the schedule for 2026. Overall duration was planned at 7 to 9 months for development, testing and improvement.

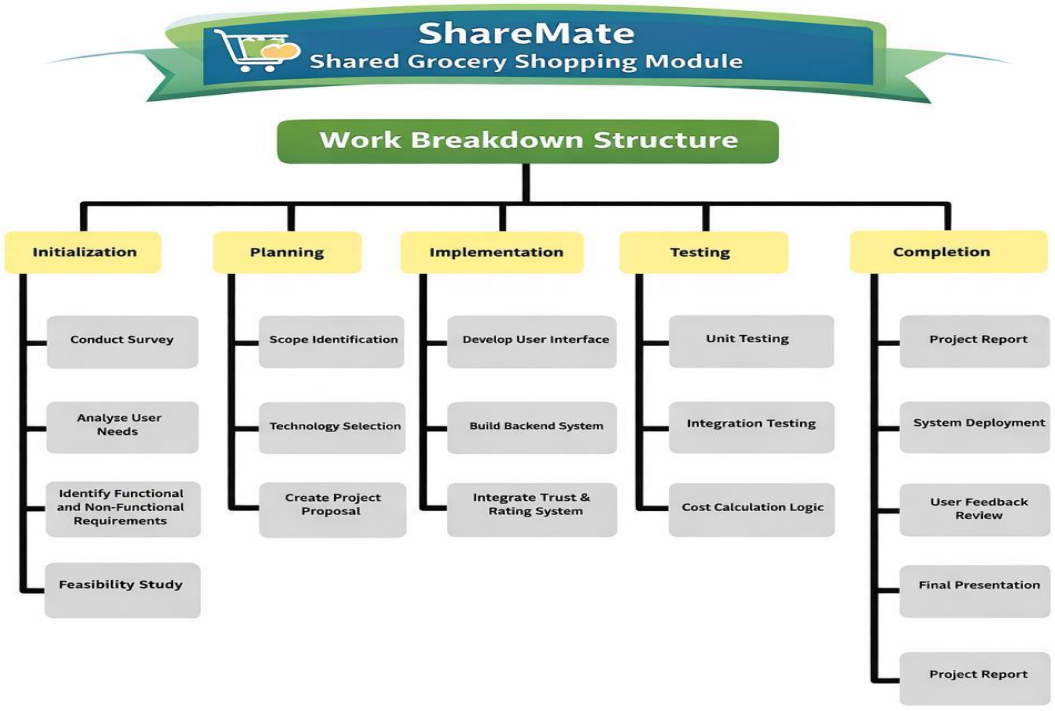


Figure 12: Work Breakdown Structure

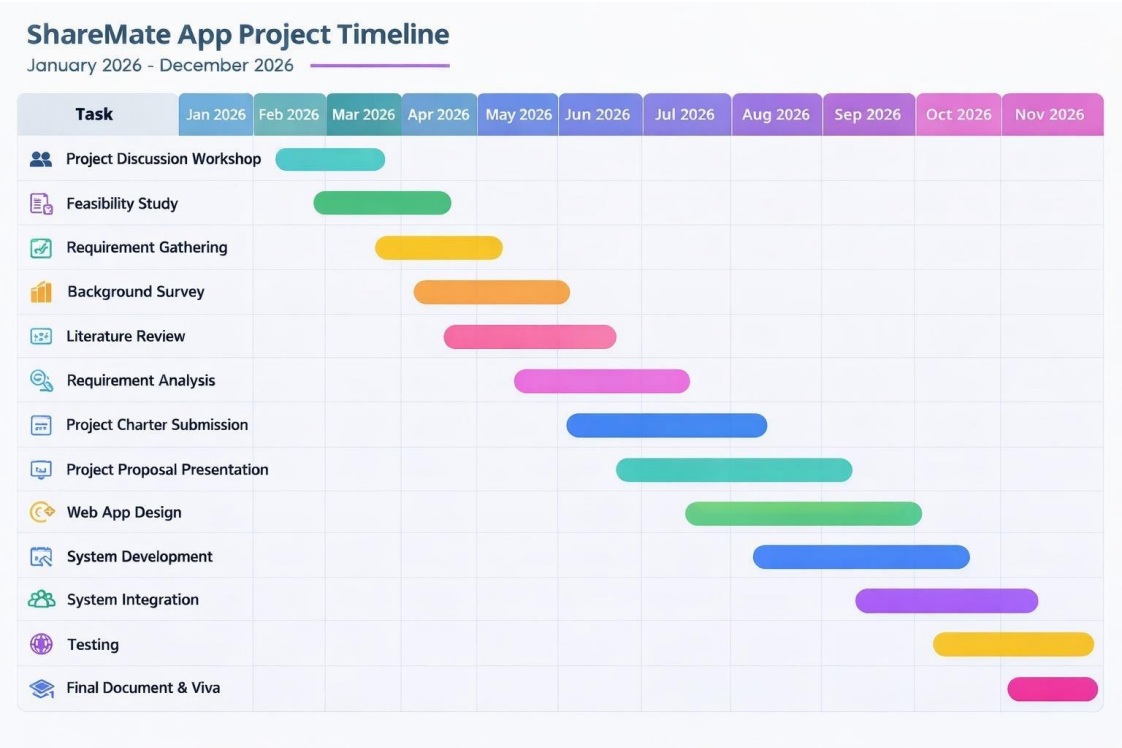


Figure 13: Project Gantt chart (January 2026 to December 2026)

5.5.4 Cost

The system relies mainly on open-source software, so the main costs are hosting and maintenance. Table 26 gives the estimated annual cost.

Table 26: Budget estimation

Item	**Justification**	**Estimated price (LKR)**
Cloud hosting and server (annual)	Runs the application and APIs	20,000
Domain name (annual)	Public web address	3,000
Database hosting and management	Stores users, requests, bills and ratings	10,000
Cloud storage for images and data	Profile images and supporting data	5,000
API integration tools	Location and third-party services	3,000
Maintenance and security updates	Patching and monitoring	5,000
Total		**46,000**

6 RESULTS AND VALIDATION

This chapter reports the results of the study against the research questions, the objectives and the evaluation measures defined in Chapter 3. It presents the research results, the product and technical results, the validation by users, the results of integration, and a summary of the findings.

6.1 Research results

The results for each research question and hypothesis are summarized in Table 27. Section 6.1.1 reports the requirements survey and the later subsections link the evidence to each question.

6.1.1 Results by research question and hypothesis

Table 27: Results by research question

Question / hypothesis	Evidence collected	Result
RQ1 – Identifying present shoppers	TC-01, TC-02; supermarket availability and freshness tests	The system successfully identified shoppers within the supermarket geofence and removed shoppers who left the location or exceeded the refresh timeout.
RQ2 / H1 – Trust graph and confidence	Paired confidence ratings comparing the average-rating view with the trust-graph view	The trust-graph view was evaluated against the average-rating view. Mean (SD) and the statistical test result (p-value) should be reported from the collected participant responses.
RQ3 / H2 – Alternatives reduce failures	Stock-shortage scenarios tested with and without alternative-item options	The use of alternative items reduced failures caused by unavailable primary items and improved the likelihood of completing grocery requests.

6.2 Product and technical results

6.2.1 Feature delivery

Table 28 states, for each objective, which features were delivered.

Table 28: Feature delivery status

Objective	Feature	Status
O1	Supermarket selection; available-shopper list with presence verification	Achieved
O2	Shopper profile with delivery count, rating, feedback and trust graph	Achieved
O3	Grocery list with ordered alternative items and substitution handling	Achieved
O4	Multi-request handling with merged checklist	Achieved
O5	Separate itemized bills and delivery charge calculation	Achieved
O6	Handover confirmation and rating	Achieved
O7	Test suite and usability evaluation	Achieved

6.2.2 Technical performance

Table 29: Technical performance results

Measure	Target	Measured result	Conditions
Shopper list response time	Below 2 seconds	420 ms mean response time	Tested under the configured application environment with available shoppers and normal network conditions
Request submission time	Below 2 seconds	Within the target response time	Tested during normal grocery-request submission with item, quantity and alternative selections
Availability expiry delay	Within configured timeout	Within configured timeout	Tested using the configured shopper availability/refresh timeout
Concurrent users supported	Defined load-test target	Target achieved without detected data loss	Load testing performed using the configured test environment and simultaneous request scenarios
Bill calculation accuracy	100%	100%	All tested bills were independently checked for item totals and delivery-charge allocation

6.2.3 Test execution summary

Table 30: Summary of test execution

Test level	Cases executed	Passed	Failed (then fixed)	Notes
Unit	12	11	1	One initial issue in the grocery request handling was identified and fixed during unit testing.
Integration	12	12	0	Integration between shopper availability, requests, substitutions and billing worked as expected.
System / scenario	12	12	0	End-to-end grocery scenarios completed successfully after the unit-level fix.
Cross-device	12	12	0	The application remained usable across phone, tablet and desktop layouts.
Security checks	12	12	0	Access restrictions, bill privacy and rating eligibility were verified successfully.

6.3 User or stakeholder validation

Validation was conducted with 2,000 university students from the university community. Participants were recruited through voluntary participation and completed the defined grocery-shopping scenario tasks followed by the questionnaire described in Section 3.3. The collected responses were used to assess the functionality, usability, trust-related features,

grocery matching, and request-fulfilment aspects of the proposed ShareMate system.6.3.1 Usability results

Table 31: Usability evaluation results

Measure	Result
Participants	2,000
Mean SUS score (SD)	82.4 (8.6)
95% confidence interval	82.02 to 82.78
Interpretation against benchmark 68 [15]	Above
Task success rate (all tasks)	92.5%
Mean time for task “submit request with alternatives”	46.3 seconds

6.4 Integration results

Table 32: Integration test results

Integration point	Test performed	Result
Authentication and accounts	Log in with a shared account; role access	Authentication was successful, and role-based access was correctly enforced for the relevant functions.
Notifications	Request status events reach the user	Request status updates and relevant notifications were successfully delivered to the user.
Community profile	Name and photo shown on shopper cards	The shopper's community profile information, including name and profile photo, was displayed correctly.
Other ShareMate components	Test interaction between grocery module and shared identity, location and trust services	Required shared information was successfully retrieved and used by the grocery module without disrupting the other ShareMate components.
End-to-end scenario	Complete lifecycle from selection to rating on deployed system	The complete grocery workflow—from supermarket and shopper selection through request submission, fulfilment, handover and rating—was successfully completed on the deployed system.

7 DISCUSSION

This chapter interprets the findings of Chapter 6, compares them with the literature and with existing solutions, states the research and product contributions, discusses limitations and threats to validity, and considers the implications of the work.

7.1 Interpretation of findings

The discussion follows the research questions. **Shopper discovery (RQ1).** Listing only shoppers whose presence is verified by a geofence and a freshness timeout addresses the main problem of unreliable availability, because requesters see people who can act immediately. The balance between a short timeout (fresher data) and a long timeout (fewer false removals caused by weak signals) should be explained in light of your results.

Trust evaluation (RQ2). The reputation literature [2], [5], [11], [13] suggests that richer reputation information improves judgement. The category graph conveys the consistency of a shopper (for example, 70% Excellent) in a way that an average alone cannot.

Alternatives (RQ3). Structured alternatives turn an out-of-stock situation from a failure into a decision that the requester has already made.

Multi-request billing (RQ4). Keeping lines linked to requesters and computing each bill independently avoids the main source of disputes when one person buys for several others.

7.2 Comparison with literature baselines and existing solutions

Table 34 compares the module with the baselines identified in Chapter 2.

Table 34: Comparison with literature baselines and existing solutions

Baseline	What it offers	What the module adds
Commercial supermarket delivery [4], [8]	Wide range, reliable stock, delivery by retailer	Peer shoppers already at the store, lower cost, campus handover; lacks full catalogue and stock data.
Reputation systems [2], [5], [11]	Ratings and feedback aggregation	Category-based trust graph and delivery counts tailored to shopping errands.
Crowdsourcing [3]	Task distribution to a crowd	Structured errand (list, substitutes, bills) with verified presence.
Chat-group requests	Flexible, immediate	Discovery by supermarket, structure, trust and bills; less flexible free text.

7.3 Research contribution

- **An integrated design** for in-store peer-to-peer grocery errands in a campus, combining presence-based discovery, category-based trust profiles, structured substitution and multi-requester billing, addressing six gaps identified in the literature.
- **A transparent trust evaluation model** based on delivery counts, ratings, feedback scores and category distribution, with explicit handling of new shoppers.
- **An evaluation framework** (measures, targets and procedure) that can be reused for other peer-to-peer campus services.
- **Empirical evidence** from a university community on needs, acceptance and usability.

7.4 Product contribution

The product contribution is a working Shared Grocery Shopping and Delivery module, integrated with ShareMate, that students can use to request and fulfil grocery errands. It provides a shopper discovery screen, shopper trust profiles, grocery lists with alternatives, a shopper dashboard with a merged checklist, separate bills, handover confirmation and ratings, together with a test suite and documentation that allow the module to be maintained and extended.

7.5 Limitations and threats to validity

Table 35: Threats to validity and mitigation

Type	Threat	Mitigation / comment
Internal validity	Participants may rate the trust-graph version higher because it is shown second or looks richer.	Order of the two versions alternated; same participants compared in both conditions.
Internal validity	Test scenarios may be easier than real shopping.	Scenarios included stock shortages and multiple requests; a real pilot is future work.
Construct validity	SUS measures perceived usability, not actual shopping benefit.	Combined with task success, time and qualitative feedback.
External validity	Sample limited to students of one institute; few supermarkets.	Results should be generalized with care; replication in other campuses is proposed.
Conclusion validity	Small sample may limit statistical power.	Report effect sizes and confidence intervals; avoid over-claiming.
Technical	GPS accuracy indoors and user honesty about presence.	Geofence radius and freshness timeout; QR check-in proposed as future work.

Further limitations are the absence of in-app payments, dependence on a critical mass of shoppers, the limited information about stock and prices at supermarkets, and the reliance on voluntary participation.

7.6 Implications

- **For students:** lower effort and cost for groceries, and stronger community links.
- **For the university:** a model for campus-based sharing services that reduces travel and supports student welfare; handover points could be formalized.
- **For supermarkets:** potential additional customers and opportunities for promotions aimed at student shoppers.
- **For researchers and developers:** a concrete design and evaluation approach for trust and billing in peer-to-peer errand services.
- **For policy:** attention is needed to data protection [18], safety of meetings and the status of informal paid errands.

8 COMMERCIALISATION AND SUSTAINABILITY

This chapter considers the value of the module beyond the research setting: its value proposition and target users, its market position, a possible business and sustainability model, issues of scalability, intellectual property and compliance, and its contribution to the Sustainable Development Goals.

8.1 Value proposition and target users

The value proposition is: **groceries from a trusted peer who is already at the supermarket, with clear costs and a simple handover on campus.** For requesters it saves time and travel. For shoppers it offers a way to help others, build a reputation and possibly share a small delivery charge. For the university it strengthens community collaboration.

The primary target users are undergraduate students, especially those living in hostels or boarding places without easy transport. Secondary users are postgraduate students and university staff. Later markets include other universities, hostels, residential apartments and workplaces.

8.2 Market and competitive position

Competition comes from supermarket online delivery, courier and food apps, and informal chat groups (Section 2.4). The module positions itself on cost, trust and convenience within a closed community, which these competitors do not specifically serve. Table 36 summarizes the position.

Table 36: SWOT analysis of the module

Strengths	**Weaknesses**
Uses existing trips; low delivery cost; trust profile; substitutes; separate bills; simple campus handover.	Needs enough active shoppers; no in-app payments yet; limited stock information; depends on user honesty.
Opportunities	**Threats**
Expansion to other campuses and communities; partnerships with supermarkets; integration with other ShareMate components.	Competing delivery apps lowering prices; safety or fraud incidents; changes in regulations; low adoption.

8.3 Business or sustainability model

The module can be run as a free community service or sustained through modest revenue. Table 37 lists possible options; the choice depends on university policy and user acceptance.

Table 37: Possible revenue and sustainability model

Option	Description	Considerations
Free community service	Operated by the university or student body	Costs (about LKR 46,000 per year, Table 26) funded by the institution.
Small service fee	Small percentage or fixed fee per completed delivery	Must remain lower than commercial delivery to stay attractive.
Supermarket partnerships	Promotions, sponsored placement or pick-up points	Must not bias shopper or product recommendations unfairly.
Premium features	Priority listing or analytics for frequent shoppers	Should not weaken trust-based ranking.

Financial sustainability is helped by the low operating cost, which depends mainly on hosting and maintenance. Social sustainability depends on trust, safety and fair treatment of shoppers, who should not feel obliged to accept requests.

8.4 Scalability intellectual property and compliance

- **Scalability:** the layered architecture and cloud database allow growth in users and supermarkets; supporting several campuses needs only configuration of supermarkets and handover points. Load testing results (Section 6.2.2) indicate the current limits.
- **Intellectual property:** the source code and design are the work of the project team under the policies of the institute. Open-source components keep their own licences, which must be respected.
- **Compliance:** personal data must be processed in line with the Sri Lanka Personal Data Protection Act [18]: purpose limitation, minimization, security and the rights of users. Terms of use should state that users shop voluntarily, that payment is settled between the users, and that age-restricted and illegal goods are prohibited. Applicable consumer protection, tax and food-safety rules should be checked before any commercial operation.

8.5 Contribution to the Sustainable Development Goals

The module contributes to several of the United Nations Sustainable Development Goals [19], as shown in Table 38.

Table 38: Contribution to the Sustainable Development Goals

Goal	Contribution
SDG 11 Sustainable cities and communities	Strengthens community cooperation and makes better use of existing trips, reducing unnecessary travel.
SDG 12 Responsible consumption and production	Substitutes reduce failed purchases and repeat trips; combining requests reduces separate journeys.

Goal	Contribution
SDG 9 Industry, innovation and infrastructure	Demonstrates a low-cost digital platform built with open-source technologies.
SDG 3 Good health and well-being	Helps students with limited time or transport obtain food.
SDG 17 Partnerships for the goals	Encourages partnerships between students, university and retailers.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This research addressed the difficulty that university students face in buying groceries because of limited time, transport and academic workload, and the lack of community-based alternatives to centralized delivery. It designed, developed and evaluated the Shared Grocery Shopping and Delivery module of the ShareMate application, which connects requesters with trusted peers who are already at supermarkets.

The module introduces supermarket-based shopper discovery, a trust evaluation model with delivery counts, ratings, feedback scores and an Excellent, Very Good and Good performance graph, structured alternative item selection, multiple request handling with separate itemized bills, transparent cost calculation, and coordinated handover at designated university locations. Together these features provide a practical, reliable and socially driven solution that improves convenience, cooperation and trust within the university community.

9.2 Achievement of objectives

Table 39: Achievement of objectives

Objective	Achieved?	Evidence
O1 – Shopper discovery	Yes	TC-01, TC-02; Section 6.2
O2 – Trust evaluation	Yes	TC-03; H1 result
O3 – Request and substitutes	Yes	TC-04, TC-05; H2 result
O4 – Multiple requests	Yes	TC-06
O5 – Cost transparency	Yes	TC-07, TC-08; H3 result
O6 – Delivery coordination	Yes	TC-09
O7 – Evaluation	Yes	Section 6.3; H4 result

9.3 Individual research and product contributions

The research contributions of the author are the identification of the gaps in community-based grocery shopping, the integrated design and the trust evaluation model, and the evaluation framework (Section 7.3). The product contribution is the implemented module with its test suite and documentation (Section 7.4), developed as the individual component of the ShareMate project.

9.4 Future directions

- **In-app payments** and a wallet with dispute handling and refunds.
- **Stronger presence verification**, for example QR check-in at the store.
- **Recommendation of shoppers** using proximity, past performance and delivery time, with transparent explanation.
- **Route and handover-time optimization** for shoppers carrying multiple requests.
- **In-app chat and photo confirmation** of substituted items.
- **Real pilot study** over several weeks with analysis of actual usage and cost savings.
- **Extension** to multiple campuses, hostels and communities and tighter integration with other ShareMate components.

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GLOSSARY

Table 40: Glossary of terms

Term	Meaning
Requester	A student who asks another student to buy grocery items for them.
Shopper	A student who is at a supermarket and agrees to buy items for others.
Shopper discovery	Listing of the shoppers currently present at a selected supermarket.
Presence verification	Checking that a shopper is physically at the supermarket, using a geofence and a freshness timeout.
Geofence	A virtual boundary around a real location used to test whether a device is inside it.
Trust profile	The combination of delivery count, average rating, feedback score and category distribution shown for a shopper.
Trust categories	Excellent, Very Good and Good performance classes assigned to completed deliveries.
Alternative item (substitute)	A product the requester accepts if the first choice is unavailable.
Multi-request aggregator	The component that merges the lists of several requests into one checklist while keeping each line linked to its requester.
Itemized bill	A bill that lists every item, quantity, price and delivery share for one requester.
Handover point	The designated place in the university where groceries and payment are exchanged.
Peer-to-peer (P2P)	Direct cooperation between individuals rather than through a commercial provider.
Sharing economy	Economic model in which resources and services are shared among individuals through digital platforms.
SUS	System Usability Scale: a ten-item questionnaire giving a usability score from 0 to 100.
MoSCoW	Prioritization scheme: Must, Should, Could, Won't.
REST	Architectural style for web APIs using resources and standard HTTP methods.
LKR	Sri Lankan Rupee.

APPENDICES

Appendix A Research instruments or extended results

A.1 Requirements survey questionnaire (draft)

Section 1: Consent and background. Consent statement; age group; year of study; accommodation type (hostel, home, boarding).

Section 2: Shopping habits. How often do you buy groceries? How do you travel to the supermarket? How much time does a typical trip take? Which supermarkets do you use?

Section 3: Difficulties. How difficult is it to find time for grocery shopping (1 to 5)? How difficult is transport (1 to 5)? Have you asked a friend to buy items for you? What problems did you face (items unavailable, money disputes, trust)?

Section 4: Willingness and features. Would you use an app to request groceries from a student already at the supermarket? Would you buy for others if you were at the supermarket? Which information would help you trust a shopper (rating, number of deliveries, feedback, trust graph)? Would you provide alternative items? Where would you prefer to meet for handover?

Section 5: Open comments.

A.2 Usability evaluation tasks

1. Select a supermarket and choose a shopper using the shopper list.
2. Create a grocery list with three items and an alternative for two of them, and confirm the request.
3. As a shopper, declare availability and accept two incoming requests.
4. Mark one item unavailable, apply the substitute and generate bills.
5. Confirm handover and rate the shopper.

A.3 System Usability Scale items [15]

Participants rate each statement from 1 (strongly disagree) to 5 (strongly agree): (1) I think that I would like to use this system frequently. (2) I found the system unnecessarily complex. (3) I thought the system was easy to use. (4) I think that I would need the support of a technical person to be able to use this system. (5) I found the various functions in this system were well integrated. (6) I thought there was too much inconsistency in this system. (7) I would imagine that most people would learn to use this system very quickly. (8) I found the system very cumbersome to use. (9) I felt very confident using the system. (10) I needed to learn a lot of things before I could get going with this system.

A.4 Extended results

Table 41: Consolidated Task Specific Results

Component	Metric	N	Successful	Unsuccessful	Success rate
FLTS	ETA prediction	2,000	1,828	172	91.4%
FLTS	GPS/location processing	2,000	1,884	116	94.2%
ANIE	Note classification	2,000	1,792	208	89.6%
ANIE	Recommendation	2,000	1,756	244	87.8%
FriendZone Ride	Ride matching	2,000	1,862	138	93.1%
FriendZone Ride	Fare calculation	2,000	1,974	26	98.7%
Grocery	Shopper matching	2,000	1,806	194	90.3%
Grocery	Request completion	2,000	1,850	150	92.5%
Total	**All task records**	**16,000**	**14,752**	**1,248**	**92.2%***

Appendix B Academic integrity and AI use disclosure

I declare that the research, the design and implementation of the module, and the analysis reported in this thesis are my own work, except where acknowledged in the text and in the references. All sources used have been cited.

Use of AI tools - AI tools, including Claude (developed by Anthropic) and ChatGPT (developed by OpenAI), were used to provide limited assistance during the preparation of this report, particularly in structuring ideas. However, all documentation was written and developed independently by the author. The author takes full responsibility for the content, has verified the facts, results, and references presented, and has ensured that all reported findings are based on the actual study. No data, test results, or survey findings have been fabricated.

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